Family Environment and Family Structure in Shaping Children's Mental Well-Being

A Study of Typically Developing And ASD Children in Urban Pabna

Statistical Field Survey Report

Department of Statistics

Pabna University of Science and Technology

Declaration

We therefore announce that the research report titled **"Family Environment and Family Structure in Shaping Children's Mental Well-Being: A Study of Typically Developing And ASD Children in Urban Pabna"** is the outcome of our original group research work.

The data for this study were gathered using a structured survey of parents and primary caregivers of children aged 6 to 12 years in metropolitan Pabna, Bangladesh. Our crew performed all data processing, statistical analyses, tables, and figures utilizing the survey dataset and statistical software.

We further certify that this report has never been presented before, either in whole or in part, for any degree, diploma, or academic award at this or any other institution. All sources of information used in this report have been properly acknowledged.

Signatures of the Survey Group Members

1 ______________________________ Roll: _________________________

2 ______________________________ Roll: _________________________

3 ______________________________ Roll: _________________________

4 ______________________________ Roll: _________________________

5 ______________________________ Roll: _________________________

Date: ________________________

Pabna University of Science and Technology

Acknowledgment

First, we express our gratitude to Allah for granting us the courage, direction, and chance to do this research.

We are incredibly appreciative of our esteemed supervisor, Dr. Md. Shamim Reza, Professor, Department of Statistics, Pabna University of Science and Technology, for his valuable guidance, unwavering support, and encouragement during the course of the study.

We further express our gratitude to Dr. Sabba Ruhi, Most. Mira Khatun, and the entire Department of Statistics for their logistical and academic assistance. The cooperation of the university administration during this investigation is greatly appreciated.

Finally, heartfelt gratitude goes to our parents, friends, and well-wishers for their constant encouragement and support.

Certificate

This is to certify that the research report entitled **"Family Environment and Family Structure in Shaping Children's Mental Well-Being: A Study of Typically Developing And ASD Children in Urban Pabna"** has been prepared by the research group under my academic supervision.

The report follows the required format and standards for undergraduate research reporting in the Department of Statistics, Pabna University of Science and Technology. I recommend this report for evaluation.

Supervisor's Signature: ___________________________

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Abstract

This study explored how family environment, family structure, screen exposure, and selected socio-demographic factors are related to children's mental well-being in urban Pabna, Bangladesh. Data were collected from 339 parents and primary caregivers of children aged 6–12 years. Among them, 254 children were typically developing and 85 children had Autism Spectrum Disorder (ASD). The survey data were cleaned and analysed using Python through descriptive statistics, reliability testing, normality tests, correlation analysis, chi-square tests, group comparison tests, regression analysis, PCA, mediation analysis, and additional screen-exposure analysis.

The overall study shows that the quality of the family environment plays an important role in children's mental well-being. Children from more supportive, calm, and emotionally safe family environments tended to show better mental well-being. ASD status was found to be the strongest factor that differentiated children's mental development outcomes. However, family structure alone did not show a significant effect after considering family environment and child group. Screen exposure also provided useful information, especially in relation to longer screen time and the timing of screen use.

Overall, our findings suggest that what matters most is not only the type of family a child lives in, but how that family functions in everyday life. A caring home environment, emotional safety, quality time with parents, and learning support appear to be more important for children's mental well-being than whether the family is nuclear, joint, or single-parent.

Keywords: family environment, family structure, mental well-being, autism spectrum disorder, screen exposure, urban Pabna, Bangladesh.

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Chapter 1: Introduction

1.1 Overview

Children's mental well-being is closely connected with the environment in which they grow up. A child's family is the first place where they learn communication, behaviour, emotional control, and social interaction. Therefore, the quality of the home environment can play an important role in shaping a child's mental and developmental outcomes.

Family environment includes factors such as emotional safety, calmness at home, parent-child quality time, family support, and learning encouragement. At the same time, family structure, such as nuclear, joint, or single-parent family, may influence how care and responsibilities are shared within the household. However, the type of family alone may not fully explain a child's well-being; how the family functions in daily life may be more important.

The main purpose of this study is to understand whether children's mental well-being is influenced more by the structure of the family or by the quality of emotional care, support, and learning environment they receive at home..

1.2 Background of the Study

In urban Bangladesh, children grow up in different family structures, such as nuclear, joint, and single-parent families. However, children's well-being may depend more on emotional safety, family support, and quality relationships than on family type alone.

This study focuses on both typically developing (TD) children and children with ASD aged 6–12 years in urban Pabna to examine how family environment, family structure, screen exposure, and related factors are associated with children's mental well-being..

1.3 Statement of the Problem.

Children's mental well-being can be influenced by how family members care, support, communicate, and spend time with them. However, family is often judged by its type, such as nuclear, joint, or single-parent, while the quality of care inside the family is overlooked.

In urban Pabna, limited research has focused on this issue among both TD children and children with ASD. Therefore, this study explores whether children's mental well-being is more related to family type or to the emotional care and support they receive at home.

1.4 Research Objectives

General Objective

To examine how family structure and family environment are related to children's mental development, comparing typically developing children and children with ASD.

Specific Objectives

1) To describe the family environment of children in urban Pabna.
2) To identify the main family structures: nuclear, joint, and single-parent.
3) To assess children's mental development from parent-reported information.
4) To examine the relationship between family environment and mental development.
5) To compare mental development across different family structures.
6) To compare mental development between typically developing children and children with ASD.
7) To understand the role of family support in children's mental development.
8) To observe the role of screen exposure and selected socio-demographic factors in children's mental well-being.

1.5 Significance of the Study

This study is important because it helps parents, teachers, health professionals, and local planners understand that children's well-being depends more on supportive family care than on family type alone. The findings may also help improve family and educational support for both typically developing children and children with ASD in urban Pabna.

1.6 Research Questions

This study was guided by the following research questions:

1) What are the major family structures among children in urban Pabna?
2) What is the overall condition of the family environment among the respondents?
3) How is family environment related to children's mental well-being?
4) Does children's mental well-being differ across nuclear, joint, and single-parent families?
5) How does mental well-being differ between typically developing children and children with ASD?
6) What role does family support play in children's mental development?

7) How are screen exposure and selected background factors associated with children's mental well-being?
8) Which factors appear to be most strongly associated with children's mental well-being?
9) Does family environment explain children's mental well-being better than family structure?

1.7 Research Limitations

The study has the following limitations:

1) The study was conducted only in urban Pabna, so the findings may not represent all children in Bangladesh.
2) Data were collected from parents or caregivers, so responses may depend on their memory and personal understanding.
3) Respondents with more than one child were asked to provide information for only one child.
4) The number of children with ASD and single-parent families was comparatively small.
5) The study shows associations among variables but does not prove cause-and-effect relationships.
6) Children's mental well-being was measured from parent-reported responses, not clinical assessment.

Chapter 2: Literature Review

2.1 Introduction

This chapter reviews literature that relates to family environment, family structure, children's mental well-being, Autism Spectrum Disorder (ASD) and screen exposure. It helps to understand the background of the study and the gap which this research covers.

2.2 Family Environment and Children's Mental Well-Being

Family environment has a strong influence on children's emotional, social, and behavioural development. A calm, caring, and supportive home can help children feel safe, confident, and encouraged. Quality time, emotional safety, and learning support from family members may improve children's mental well-being.

2.3 Family Structure and Child Development

Children may grow up in nuclear, joint, or single-parent families. Each family structure may provide different levels of care, support, and responsibility-sharing. However, family type alone may not determine children's well-being. The quality of care and relationships within the family may be more important.

2.4 Typically Devoloped Children and Children with ASD

Typically developing children generally follow expected patterns of growth, communication, and behaviour. Children with ASD may face more challenges in communication, social interaction, attention, and adjustment. For them, family support, patience, routine, and emotional care are especially important.

2.5 Screen Exposure and Children's Well-Being

Screen use has become common in children's daily life. Guided and limited screen use may support learning, but excessive screen time may affect sleep, attention, behaviour, and social interaction. Therefore, screen exposure is an important related factor in children's mental well-being.

2.6 Research Gap

Although many studies discuss family and child development, limited research has focused on children in urban Pabna. There is also limited evidence comparing typically developing children and children with ASD in relation to family environment, family structure, and screen exposure.

This study addresses this gap by examining whether children's mental well-being is more related to family type or to the quality of care and support they receive at home.

Chapter 3: Research Methodology

3.1 Introduction

The study's methodology is covered in this chapter..It includes the research design, study area, target population, sampling technique, sample size, data collection procedure, study variables, data processing, statistical analysis, and ethical considerations.

3.2 Study Design

The survey used a cross-sectional descriptive and analytical design. This methodology is suitable for the exploratory and comparative goals of the current study since it allows for the simultaneous analysis of several variables and their correlations at a single moment in time.

3.3 Population

The study was conducted in urban Pabna, a district town in the Rajshahi Division of Bangladesh. Pabna was selected because of its representative mix of family types—nuclear, joint, and single-parent—as well as the availability of both regular and special educational institutions serving children with ASD.

3.4 Sampling Technique

A purposive sampling technique was used to select respondents. This method was appropriate because the study required responses from parents and caregivers of specific child groups, especially children with ASD, who are not always easy to reach through random sampling.

3.5 Sample Size Determination

The sample size for this study was determined using the formula for comparing two independent proportions in a cross-sectional study. In this study, children with ASD were considered the exposed group, while typically developing children were considered the unexposed group.

The expected prevalence for the unexposed group was taken from the National Mental Health Survey Bangladesh 2018-2019. The survey reported that the prevalence of mental disorders among children aged 7-12 years was 12.4%, with a 95% confidence interval of 9.8% to 15.0%. For this study, the upper limit of this confidence interval, 15.0%, was used as the expected prevalence among TD children. This was done to obtain a conservative sample size estimate. The expected prevalence among children with ASD was assumed to be 30.0%. giving a risk/prevalence ratio of about 2.0. The calculation was performed using OpenEpi Version 3 with a 95% confidence level, 80% power, and an unexposed-to-exposed ratio of 3:1.

The calculation followed the two-proportion sample size approach for independent groups. This method was appropriate because the study compared two separate child groups using expected outcome proportions. Sharma et al. (2020) noted that sample size selection should be based on the study design, confidence level, study power, expected event rate, effect size, and group ratio..

Table 6. Prevalence of mental disorders (diagnosed following DSM 5*) in children

Age group (years)	Boy		Girl		Urban		Rural		All	
	%	95% CI*	%	95% CI*	%	95% CI*	%	95% CI*	%	95% CI*
7 - 12	13.2	9.8 - 16.7	11.6	8.3 - 14.9	12.5	8.8 - 16.2	12.4	9.2 - 15.5	12.4	9.8 - 15.0
13 - 17	14.3	9.8 - 18.8	11.5	7.7 - 15.2	10.3	6.1 - 14.4	13.6	9.8 - 17.3	12.8	9.8 - 15.9
7 - 17	**13.7**	**10.9 - 16.5**	**11.5**	**8.8 - 14.3**	**11.5**	**8.8 - 14.2**	**12.9**	**10.3 - 15.5**	**12.6**	**10.5 - 14.7**

* American Psychiatric Association. Diagnostic and Statistical Manual of Mental Disorders (DSM–5). 2013
† Confidence interval

Figure 1: Reference Prevalence Used for Sample Size Determination (National Mental Health Survey 2019)

The marked values show the prevalence of mental disorders among children aged 7-12 years reported in the National Mental Health Survey Bangladesh 2018-2019. The upper limit of the 95% confidence interval, 15.0%, was used as the expected prevalence for the unexposed group.

The assumptions used for sample size calculation are shown below:

Parameter	Value
Confidence level	95%
Study power	80%
Unexposed-to-exposed ratio	3:1
Expected prevalence in unexposed group (TD), p_0	0.15
Expected prevalence in exposed group (ASD), p_1	0.30
Standard normal value for 95% confidence, $Z_{1\text{-alpha}/2}$	1.96
Standard normal value for 80% power, $Z_{1\text{-beta}}$	0.84

Table 1: Parameters Used for Sample Size Calculation

Kelsey Method

The Kelsey formula for the exposed group sample size is:

$$n_1 = \frac{\left[Z_{1-\alpha/2}\sqrt{(r+1)p^{-}(1-p^{-})} + Z_{1-\beta}\sqrt{rp_0(1-p_0)+p_1(1-p_1)}\right]^2}{r(p_1-p_0)^2}$$

where,

$$p^{-} = \frac{p_1 + rp_0}{r+1}$$

Substituting the values:

$$p^{-} = \frac{0.30 + 3(0.15)}{3+1} = 0.1875$$

$$n_1 = \frac{\left[1.96\sqrt{(3+1)(0.1875)(1-0.1875)} + 0.84\sqrt{3(0.15)(1-0.15)+0.30(1-0.30)}\right]^2}{3(0.30-0.15)^2}$$

$$n_1 \approx 70.19 \approx 71$$

Thus, by the Kelsey method:

$$n_{ASD} = 71$$

$$n_{TD} = 3 \times 71 = 213$$

$$Total\ sample\ size = 71 + 213 = 284$$

Fleiss Method with Continuity Correction

First, the uncorrected Fleiss sample size is calculated as:

$$n_F = \frac{\left[Z_{1-\alpha/2}\sqrt{(r+1)p^{-}(1-p^{-})} + Z_{1-\beta}\sqrt{rp_1(1-p_1)+p_0(1-p_0)}\right]^2}{r(p_1-p_0)^2}$$

Substituting the values:

$$n_F = \frac{\left[1.96\sqrt{(3+1)(0.1875)(1-0.1875)} + 0.84\sqrt{3(0.30)(1-0.30)+0.15(1-0.15)}\right]^2}{3(0.30-0.15)^2}$$

$$n_F \approx 75.74 \approx 76$$

Then the Fleiss continuity correction is applied:

$$n_{cc} = \frac{n_F}{4}\left[1 + \sqrt{1 + \frac{2(r+1)}{n_F r |p_1 - p_0|}}\right]^2$$

Substituting the values:

$$n_{cc} = \frac{75.74}{4}\left[1 + \sqrt{1 + \frac{2(3+1)}{75.74 \times 3 \times |0.30 - 0.15|}}\right]^2$$

$$n_{cc} \approx 84.40 \approx 85$$

Thus, the exposed group sample size is:

$$n_{ASD} = 85$$

Since the unexposed-to-exposed ratio was 3:1:

$$n_{TD} = 3 \times 84.40 = 253.20 \approx 254$$

Therefore:

$$Total\ sample\ size = 85 + 254 = 339$$

Final Sample Size Selection

Method	Exposed (ASD)	Unexposed (TD)	Total
Kelsey	71	213	284
Fleiss	76	228	304
Fleiss with Continuity Correction	85	254	339

Table 2: Sample Size Calculation Summary

Among the three estimates, the Fleiss method with continuity correction produced the largest and most conservative sample size. Therefore, the final sample size selected for this study was 339 children, including 85 children with ASD and 254 typically developing children. This sample size was considered suitable for comparing the two child groups and for conducting further statistical analysis.

3.6 Data Collection Procedure

Data were collected using a structured survey questionnaire. A structured questionnaire was used as the main research instrument. It included categorical and Likert-scale questions on family environment, screen exposure, and children’s mental well-being. Parents and caregivers were asked to provide information for one child aged 6–12 years. Respondents with more than one child were asked to provide information for only one child to maintain consistency in the dataset.

Figure 2: Data collection using paper based questionnaire conducted by the research group

3.7 Data Entry

The survey data were entered digitally by all of our group members using Google Forms/Google Sheets. Each entry was carefully recorded so that one response represented information about one selected child only. After data entry, the responses were combined into a single dataset and exported for analysis.

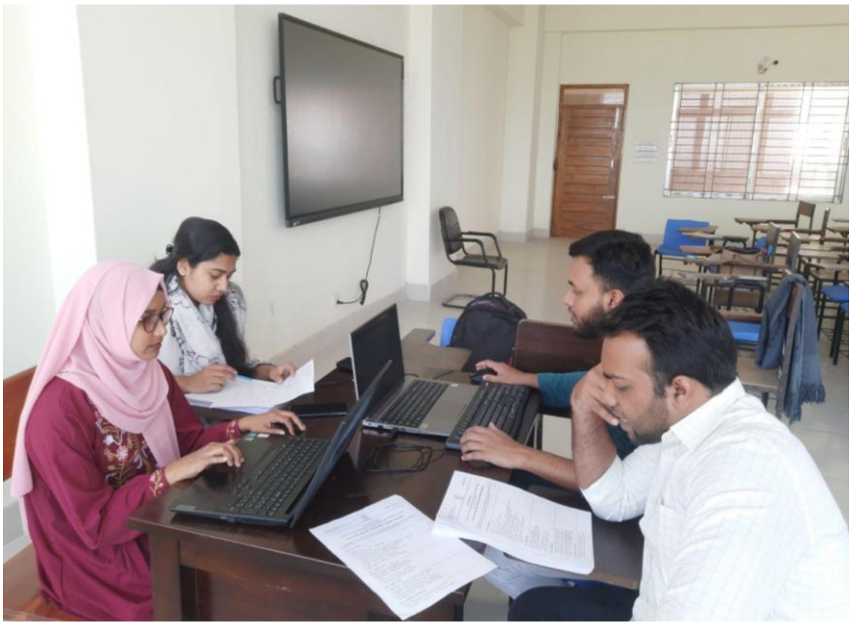


Figure 3: Collected responses were entered and organized digitally by the research group.

3.8 Data Processing

After data entry, the dataset was checked and cleaned before analysis. Missing values, duplicate records, inconsistent categories, and possible entry errors were reviewed. Categorical responses were standardized, and Likert-scale responses were converted into numerical scores.

Composite scores were created for family environment and children's mental well-being. The cleaned dataset was then prepared for statistical analysis using Python.

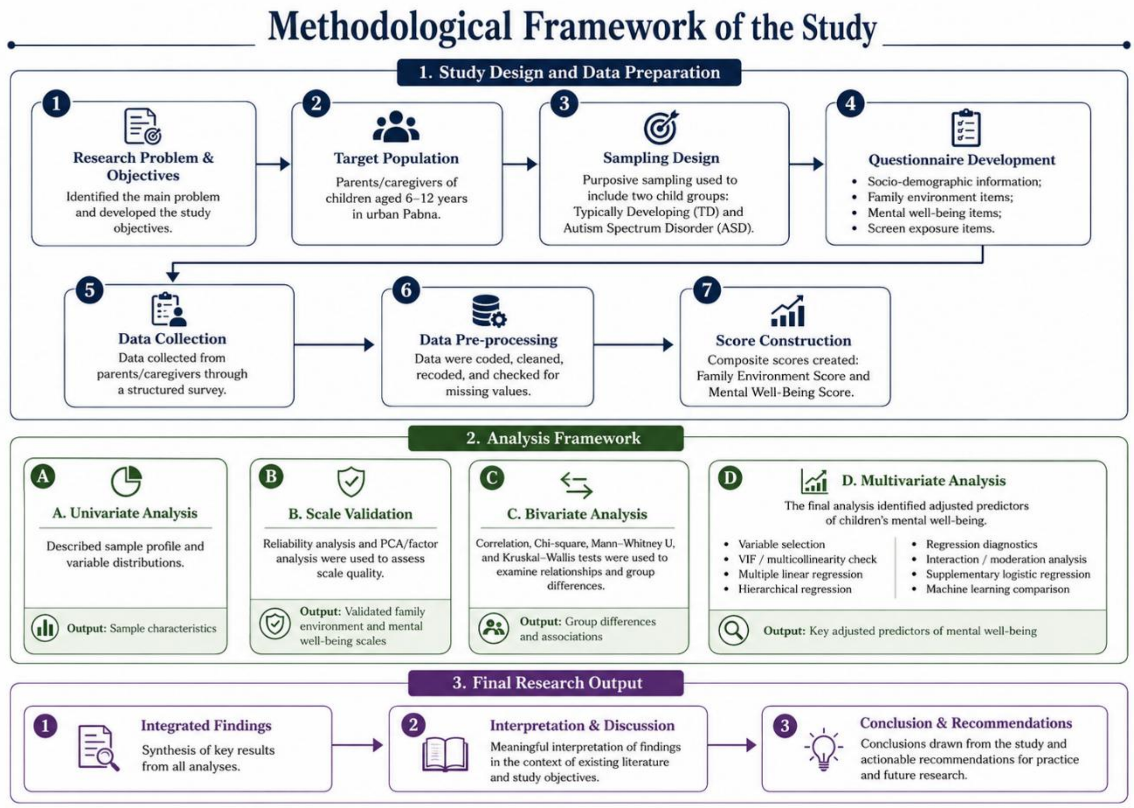
3.9 Study Variables

The main outcome variable of the study was children's mental well-being. The key explanatory variables were family environment, family structure, child group, screen exposure, and selected socio-demographic factors.

Family structure was categorized as nuclear, joint, and single-parent family. Family environment was measured through indicators such as emotional safety, home atmosphere, parent-child quality time, and learning support. Mental well-being was measured using parent-reported emotional, behavioural, social, and developmental indicators.

3.10 Statistical Analysis

The analysis was carried out in several stages to describe the data, examine relationships among variables, and identify the factors associated with children's mental well-being.



1) Univariate analysis: frequency, percentage, mean, standard deviation, and graphical summaries.

2) Bivariate analysis: cross-tabulation, chi-square test, Spearman correlation, and group comparisons.

3) Inferential analysis: normality test, Mann-Whitney U test, ANOVA, Kruskal-Wallis test, post-hoc tests, and two-way ANOVA.

4) Multivariate analysis: OLS regression, standardized coefficients, regression diagnostics, and hierarchical regression.

5) Supplementary analysis: logistic regression,mediation analysis, and ML model comparison.

These methods were used to describe the data, test relationships, compare groups, and identify factors associated with children's mental well-being.

3.11 Ethical Considerations

Participation in the survey was voluntary, and the information was collected only for academic research purposes. Respondents' personal identities were kept confidential and were not disclosed in the report.

The data were analysed and presented in summary form to protect privacy. The study respected the dignity, consent, and confidentiality of all respondents and children included in the survey.

Chapter 4: Data Analysis

This chapter presents the analysis of the collected survey data.The findings are presented through tables, figures, and brief interpretations to make the results clear and meaningful.

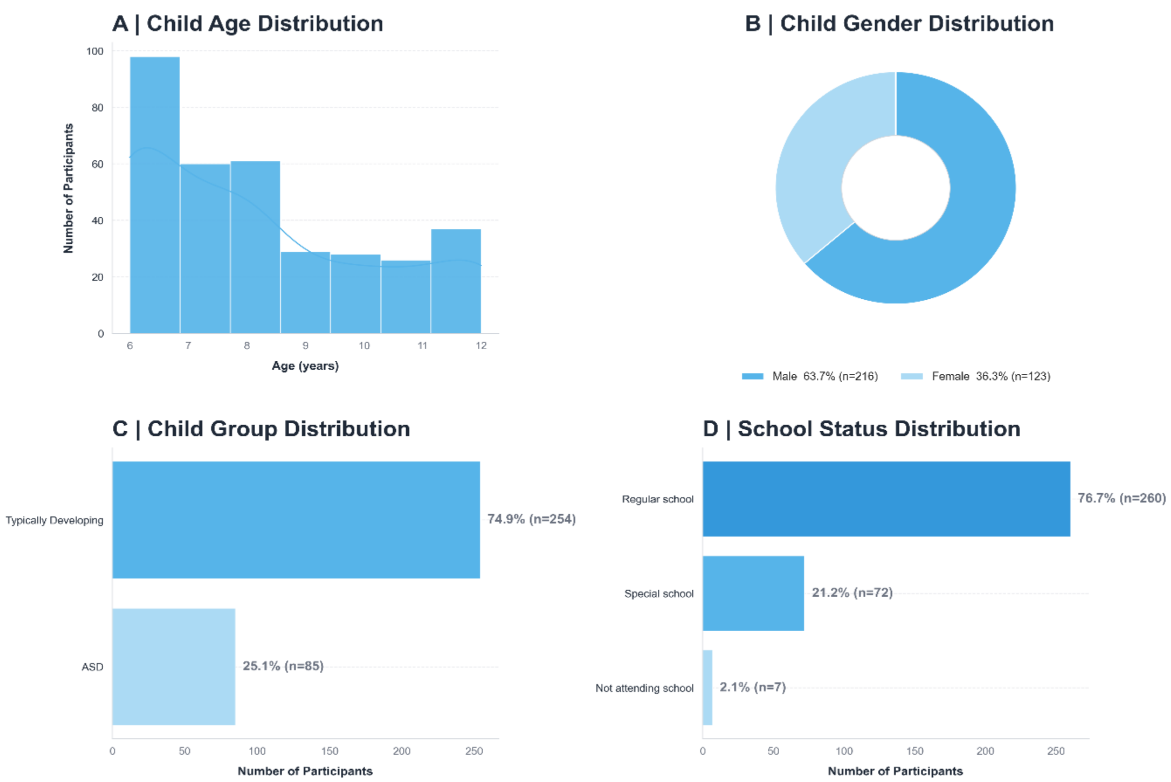
4.1 Socio-Demographic Characteristics of Respondents

Main socio-demographic table

Table 3: Socio-demographic characteristics of respondents

Characteristic	Detail	Frequency (N)	Percentage (%)
Child Age (years)	Range: 6-12; M = 8.16, SD = 2.04	339	100
Child Gender	Male	216	63.7
	Female	123	36.3
Child Group	Typically Developing	254	74.9
	ASD	85	25.1
Family Structure	Nuclear Family	217	64
	Joint Family	108	31.9
	Single-Parent Family	14	4.1

Socio-demographic visual profile



Interpretation:

Panel A: The largest number of children were aged 6 years, and all children were within the target age range of 6–12 years.

Panel B: Male children formed the majority (63.7%) compared to female children (36.3%).

Panel C: Most children were typically developing (74.9%), while 25.1% were children with ASD.

Panel D: Most children attended regular school (76.7%), followed by special school (21.2%). Very few were not attending school (2.1%).

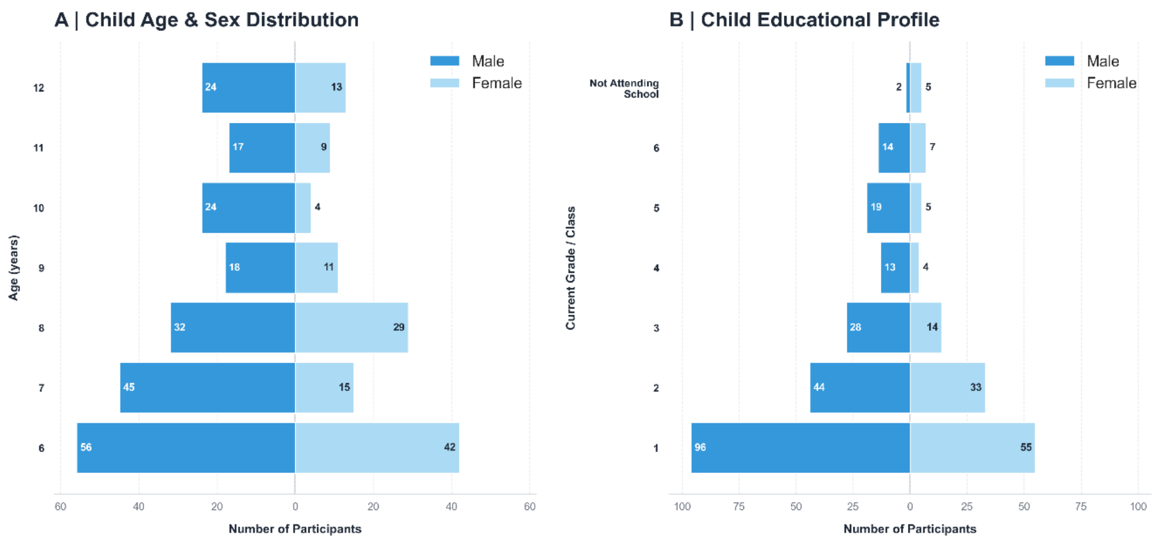


Figure 4: Socio-demographic profile: age, gender, child group, and school status

Interpretation:

Panel A: Most children were in the younger age groups, especially ages 6–8 years. Male children were higher than female children in most age categories.

Panel B: Most children were in Grade 1, followed by Grade 2 and Grade 3. Only a few children were not attending school.

Family structure and household size

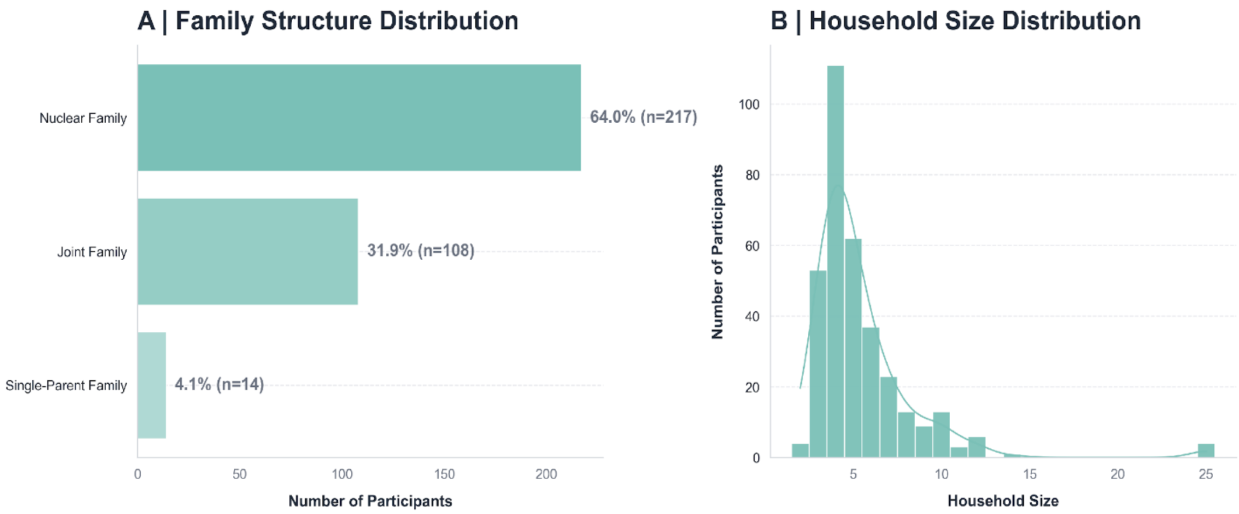


Figure 5: Family structure and household size distribution

Interpretation:

Panel A: Most children belonged to nuclear families (64.0%), followed by joint families (31.9%) and single-parent families (4.1%).

Panel B: Most households had around 4–6 members, showing that small to medium-sized families were more common.

4.2 Family Environment Score

The Family Environment Score was examined through five indicators: emotional atmosphere, argument frequency, emotional safety, parent-child quality time, and family learning support.

Table 4: Frequency distribution of emotional atmosphere

Emotional Atmosphere	Frequency	Percentage (%)
Neither stressful nor calm	125	36.9
Mostly calm	79	23.3
Very calm and supportive	75	22.1
Somewhat stressful	50	14.7
Very stressful	10	2.9

Table 5: Frequency distribution of argument frequency

Argument Frequency	Frequency	Percentage (%)
Rarely	145	42.8
Sometimes	120	35.4
Never	59	17.4
Often	12	3.5

Argument Frequency	Frequency	Percentage (%)
Very Often	3	0.9

Table 6: Frequency distribution of emotional safety

Emotional Safety	Frequency	Percentage (%)
Completely	177	52.2
Mostly	71	20.9
Moderately	58	17.1
Slightly	21	6.2
Not at all	12	3.5

Table 7: Frequency distribution of parent-child quality time

Parent Child Quality Time	Frequency	Percentage (%)
Almost every day	246	72.6
1-2 days a week	45	13.3
3-4 days a week	43	12.7
Almost never	5	1.5

Table 8: Frequency distribution of family learning support

Family Learning Support	Frequency	Percentage (%)
Strongly agree	158	46.6
Agree	136	40.1
Neutral	34	10
Disagree	11	3.2

Table 9: Mean Family Environment Score by family structure

Family Type	Mean	SD	N
Joint Family	*3.624*	0.621	108
Nuclear Family	*3.682*	0.593	217
Single-Parent Family	*3*	0.748	14

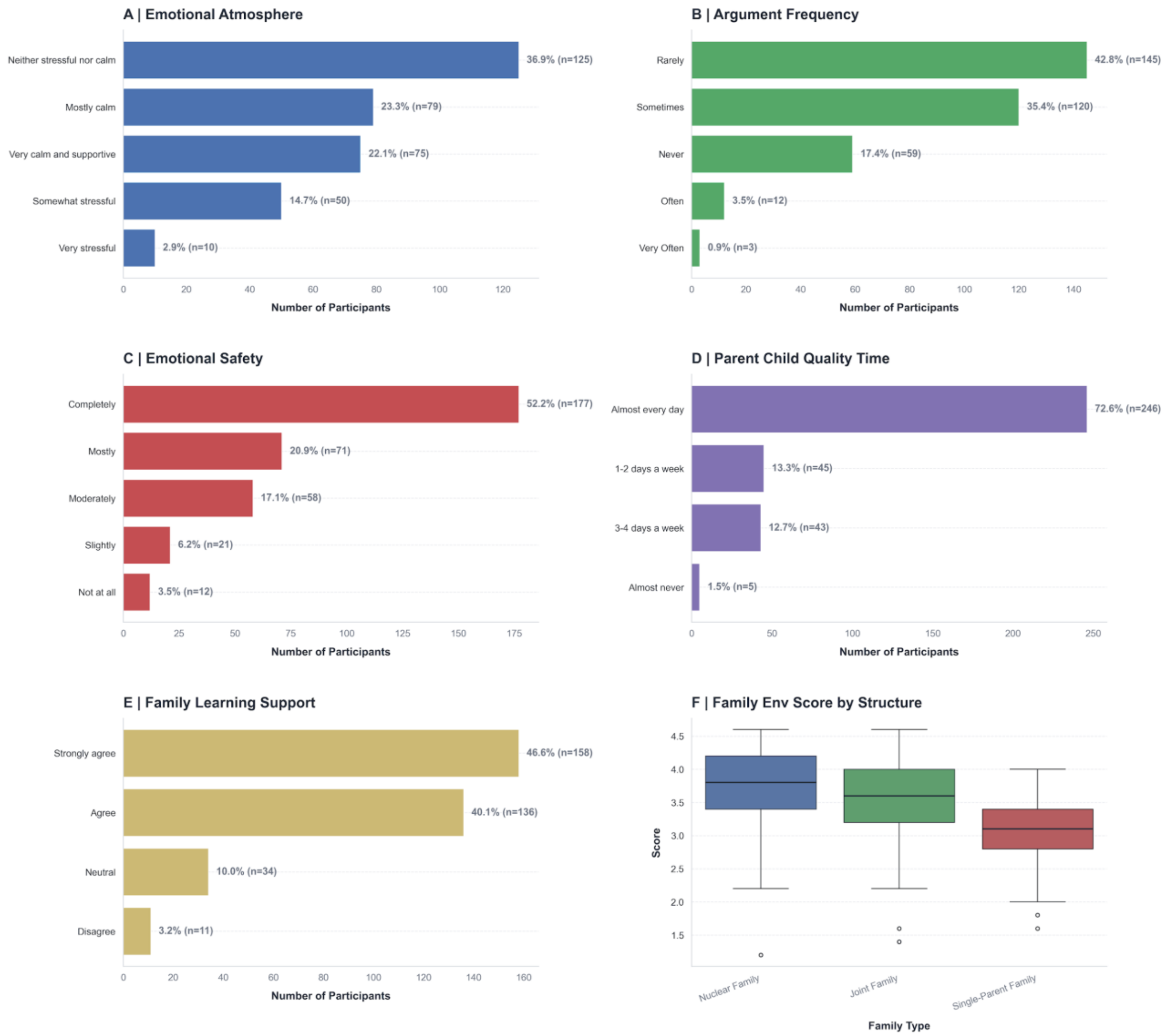


Figure 6: Family environment indicators and composite score by family structure

Interpretation:

Emotional Atmosphere : Most families reported a neutral to positive home atmosphere. The largest group described their home as neither stressful nor calm (36.9%), while 45.4% reported it as mostly calm or very calm and supportive. Only 2.9% reported a very stressful environment.

Insight: Most children lived in homes that were not highly stressful, although many families were neutral rather than strongly supportive.

Argument Frequency : Family arguments were generally low. Most respondents reported that arguments occurred rarely (42.8%) or sometimes (35.4%). Only a small proportion reported frequent arguments.

Insight: High-conflict family environments were uncommon in the sample.

Emotional Safety :Emotional safety was one of the strongest indicators. More than half of the children were reported to feel completely safe at home (52.2%), while 20.9% felt mostly safe.

Insight: Most children experienced a secure emotional environment within their families.

Parent-Child Quality Time :Most respondents reported spending quality time with their child almost every day (72.6%).

Insight: Regular parent-child interaction was common and may support children's emotional and behavioural development.

Family Learning Support : Family learning support was strong. Most respondents strongly agreed (46.6%) or agreed (40.1%) that their family supported the child's learning.

Insight: Many families were actively involved in children's educational and developmental support.

Family Environment Score by Family Structure : The mean Family Environment Score was highest among nuclear families (M = 3.682), followed by joint families (M = 3.624). Single-parent families had a lower mean score (M = 3.000).

Insight: Nuclear and joint families showed similar and relatively better family environment scores, while single-parent families appeared more vulnerable. However, this should be interpreted carefully because the single-parent group was small.

Table 10: Item-wise Family Environment Scores by family structure

Family Type	Emotional Atmosphere	Argument Frequency	Emotional Safety	Parent Child Quality Time	Family Learning Support
Joint Family	3.333	3.657	4.204	3.574	3.352
Nuclear Family	3.585	3.774	4.157	3.59	3.304
Single-Parent Family	2.714	3.429	2.929	3.071	2.857
Overall	3.469	3.723	4.121	3.563	3.301

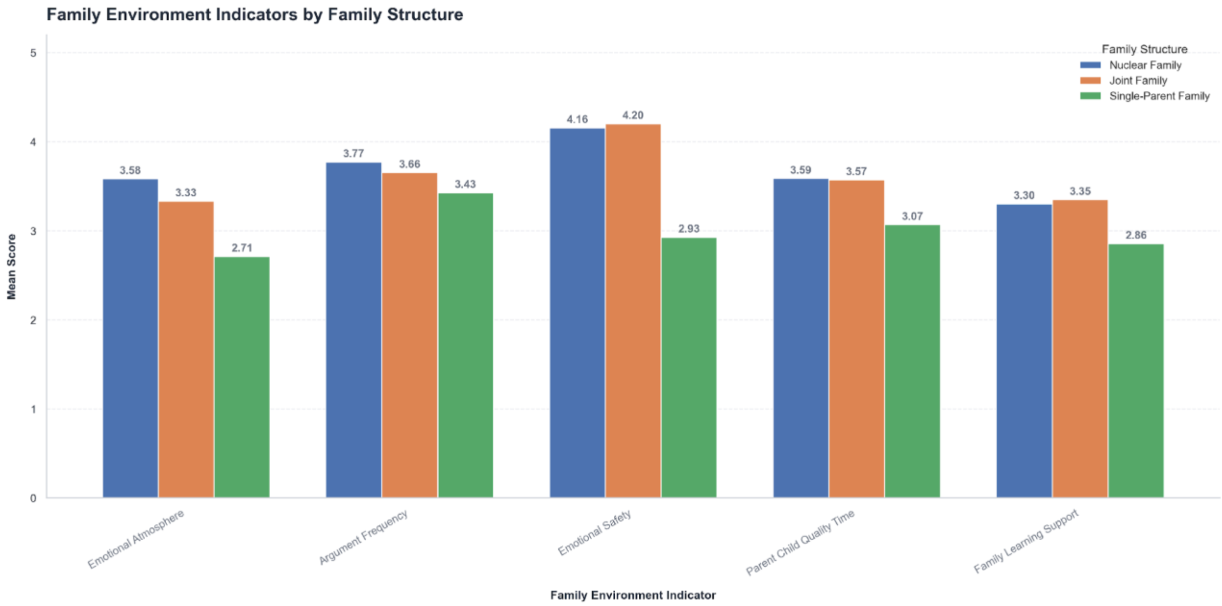


Figure 7: Item-wise family environment indicators by family structure

Interpretation

Nuclear and joint families scored higher than single-parent families in most indicators. Joint families showed the highest emotional safety score, while nuclear families showed slightly better emotional atmosphere and lower argument frequency.

Insight : The quality of family environment differed across family structures, but nuclear and joint families showed more stable patterns than single-parent families.

4.3 Mental Well-Being Score

The Mental Well-Being Score was assessed using parent-reported emotional, behavioural, social, and developmental indicators. The results show a clear difference between typically developing children and children with ASD.

Table 11: Mental Development Score by child group

Child Group	Mean	SD	Min	Max	N
ASD	2.482	0.74	1	4.5	85
Typically Developing	3.998	0.582	2	5	254

Table 12: Item-wise mental well-being profile by child group

Group Item	ASD	Typically Developing
Adapt Struggle	1.988	3.61

Group Item	ASD	Typically Developing
Behavior Problems	2.518	4.35
Difficult Behavior	2.365	3.709
Emotional Recovery	2.376	3.835
Express Needs	3.012	4.602
Focus	2.635	3.843
Social Interest	2.894	4.331
Worried Anxious	2.071	3.705

Figure 8: Mental well-being score: 4-panel analysis

Interpretation

Panel A: Overall Distribution

Panel A shows that most children had Mental Well-Being Scores between **3.5 and 4.5**, indicating a generally moderate to good level of mental well-being in the full sample.

Panel B: Score by Child Group

Panel B shows a clear difference between the two groups. Typically developing children had higher Mental Well-Being Scores, while children with ASD had lower scores and greater variation.

Panel C: Item-wise Mean Comparison

Panel C shows that typically developing children scored higher than children with ASD in all mental well-being items. The largest differences appeared in expressing needs, social interest, behavioural problems, and adaptive struggle.

Panel D: Density by Group

The two groups show different score patterns. TD children's scores were mostly higher, while ASD children's scores were mostly lower.

4.4 Screen Exposure Profile

Table 13: Screen time distribution

Screen Time	Frequency	Percentage (%)
No Screen Use	14	4.1
Less than 1 hour	87	25.7
1-2 hours	134	39.5
3-4 hours	71	20.9
More than 4 hours	33	9.7

Table 14: Screen use timing distribution

Screen Use Time	Frequency	Percentage (%)
Morning	15	4.4
Noon	73	21.5
Afternoon	69	20.4
Evening	82	24.2
Night	86	25.4
Not Applicable	14	4.1

Table 15: Screen content distribution

Screen Content	Frequency	Percentage (%)
Educational/learning	39	11.5
Entertainment (cartoons, YouTube)	170	50.1
Games	34	10
Mixed	77	22.7
Social media	5	1.5
No Screen Content	14	4.1

Table 16: Screen time by child group

Child Group Screen Time	ASD	Typically Developing
No Screen Use	1.2	5.1
Less than 1 hour	15.3	29.1
1-2 hours	36.5	40.6
3-4 hours	27.1	18.9
More than 4 hours	20	6.3

Figure 9: Screen exposure profile of the sample

Interpretation

Screen usage is nearly universal: 95.9% of children (325 out of 339) used screens, while only 4.1% reported no screen use.

Most common duration: The highest proportion of children used screens for 1–2 hours daily (39.5%), followed by less than 1 hour (25.7%). However, 30.6% used screens for 3 hours or more per day.

Peak usage time: Screen use was most common at night (25.4%) and evening (24.2%), which may indicate possible concerns related to sleep routine.

Content dominated by entertainment: Entertainment content such as cartoons and YouTube was the most common screen content (50.1%), while educational content was much lower (11.5%).

Higher screen use among ASD children: Children with ASD showed higher heavy screen use than typically developing children. More than 4 hours of screen use was reported by 20.0% of ASD children compared with 6.3% of TD children.

4.5 Parental Education Profile

Table 17: Father education level distribution

Father Education	Frequency	Percentage (%)
No formal schooling	10	2.9
Madrassa Education	2	0.6
Primary (Up to Class 5)	32	9.4
Secondary (SSC/Class 10)	58	17.1
Higher Secondary (HSC/12)	69	20.4
Diploma/Certificate	13	3.8
Bachelor's Degree	65	19.2
Master's Degree or Higher	89	26.3

Table 18: Mother education level distribution

Mother Education	Frequency	Percentage (%)
No formal schooling	12	3.5
Madrassa Education	1	0.3
Primary (Up to Class 5)	27	8
Secondary (SSC/Class 10)	73	21.5

Mother Education	Frequency	Percentage (%)
Higher Secondary (HSC/12)	72	21.2
Diploma/Certificate	13	3.8
Bachelor's Degree	69	20.4
Master's Degree or Higher	72	21.2

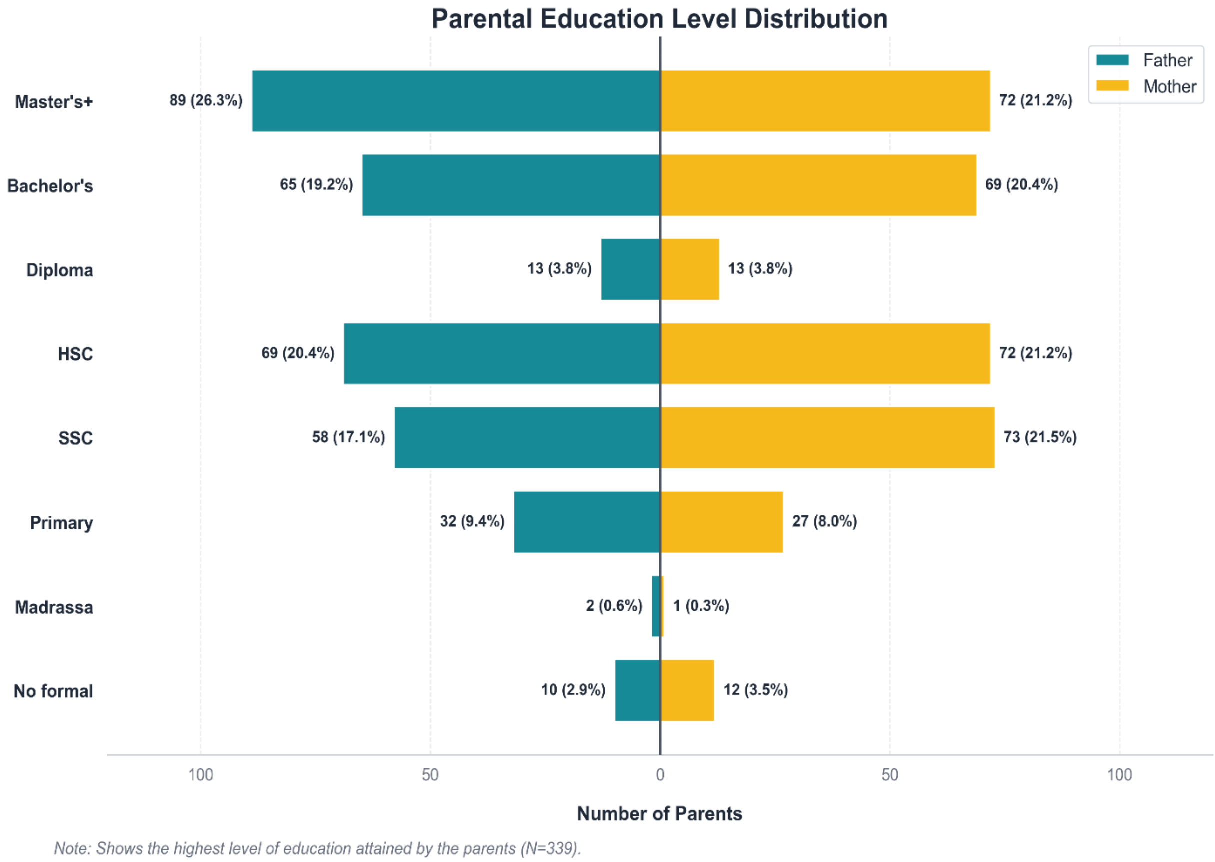


Figure 10: Parental education profile

Interpretation

Fathers had relatively high education levels: The largest group of fathers had a Master's degree or higher (**26.3%**), followed by Higher Secondary (**20.4%**) and Bachelor's degree (**19.2%**).

Mothers also had a balanced education profile: Most mothers completed Secondary (**21.5%**), Higher Secondary (**21.2%**), Master's degree or higher (**21.2%**), or Bachelor's degree (**20.4%**).

Low Literacy : Only **2.9%** of fathers and **3.5%** of mothers had no formal schooling, indicating a relatively educated sample

The parental education profile indicates that most children came from families where parents had at least secondary or higher levels of education. This may positively influence children's learning support, awareness, and family environment.

4.6 Parental Occupation Profile

Table 19: Father occupation distribution

Father Occ Clean	Frequency	Percentage (%)
Business	116	34.2
Private Service	106	31.3
Govt. Service	64	18.9
Daily Laborer	29	8.6
Immigrant	11	3.2
Farmer	5	1.5
Other	4	1.2
Unemployed	2	0.6
Immigrant	2	0.6

Table 20: Mother occupation distribution

Mother Occupation	Frequency	Percentage (%)
Homemaker	297	87.6
Private Service	18	5.3
Govt. Service	17	5
Business	5	1.5
Daily Laborer	2	0.6

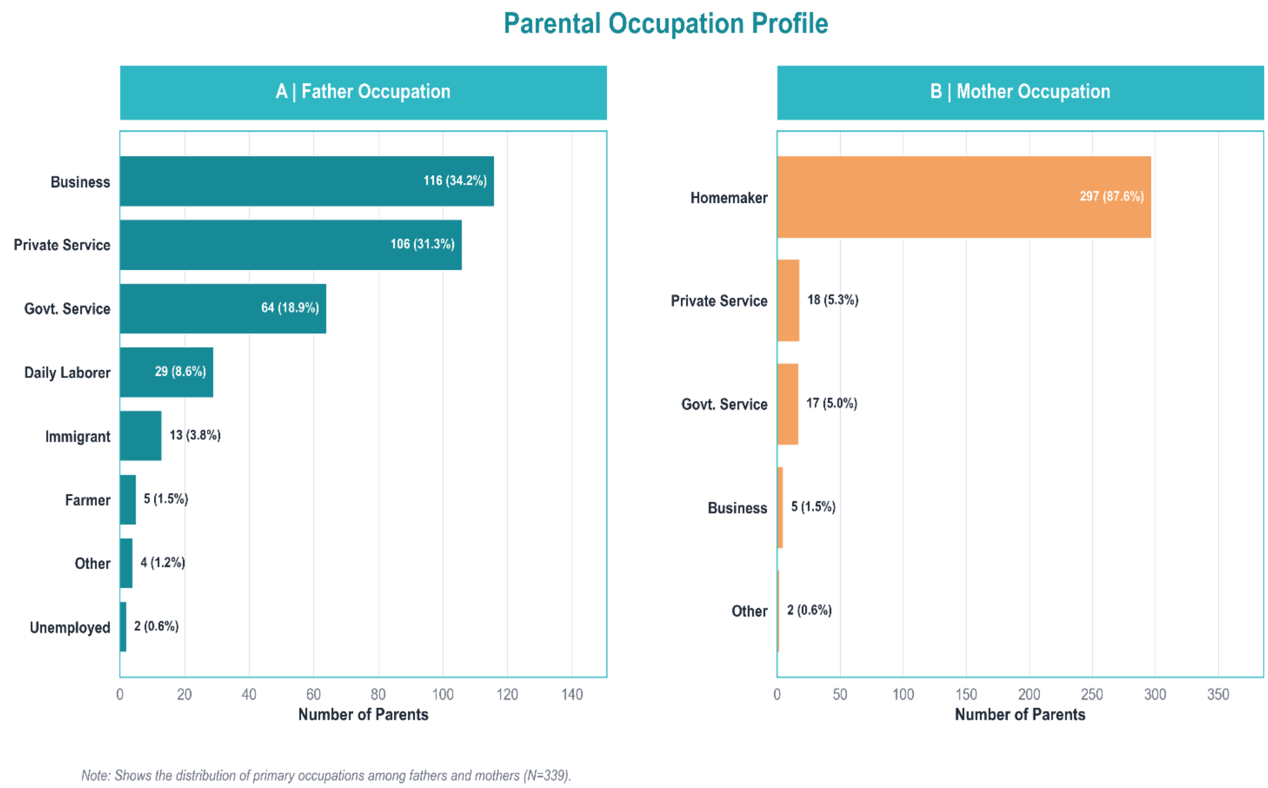


Figure 11: Parental occupation profile

Interpretation

Most fathers were engaged in **business (34.2%)**, **private service (31.3%)**, or **government service (18.9%)**, indicating that fathers were commonly the main earning members of the family. A smaller proportion worked as daily laborers or in other occupations, showing some variation in economic background.

Most mothers were **homemakers (87.6%)**, while only a small percentage were involved in paid employment. This suggests that mothers were mainly responsible for household and caregiving roles.

Overall, the occupation pattern reflects a common family arrangement where fathers mostly contributed to income and **mothers mainly supported child care and home management.**

4.7 Reliability Analysis

Reliability analysis was conducted to check whether the scale items were consistent and suitable for further analysis. Reliability analysis is the process of testing whether a research tool produces consistent results under the same conditions. It shows whether a survey, questionnaire, or test gives similar outcomes when repeated. We can also say it shows whether your results hold same when you measure the same thing more than once. Before proceeding with inferential analyses, the internal consistency of the two composite scales was evaluated using Cronbach's Alpha.

Table 21: Cronbach's alpha and scale reliability statistics

Scale	No. Items	Cronbach's Alpha	Interpretation
Family Environment Scale	5	0.693	Good
Mental Development Scale	8	0.875	Acceptable

Table 22: Mental Development Scale item-total reliability

Item	Item-Total Correlation	Alpha if Deleted
Express Needs	0.687	0.855
Worried Anxious	0.616	0.862
Social Interest	0.573	0.866
Emotional Recovery	0.625	0.861
Difficult Behavior	0.592	0.865
Focus	0.56	0.868
Adapt Struggle	0.675	0.856
Behavior Problems	0.758	0.846

Table 23: Family Environment Scale item-total reliability

Item	Item-Total Correlation	Alpha if Deleted
Emotional Atmosphere	0.579	0.58
Argument Frequency	0.392	0.667
Emotional Safety	0.461	0.645
Parent Child Quality Time	0.371	0.674
Family Learning Support	0.473	0.639

Interpretation

- Cronbach's Alpha measures the internal consistency of a composite scale. Values above **0.70** are generally acceptable, while values between **0.60 and 0.70** may be acceptable for exploratory survey research with heterogeneous items (George & Mallery, 2003).
- The Mental Development Scale showed strong reliability with a Cronbach's Alpha of 0.875, indicating that the eight items were highly consistent. The item-total

correlations were also satisfactory, suggesting that each item made a meaningful contribution to the overall scale. The Alpha-if-Deleted values did not increase the overall reliability, so all items were retained for further analysis.

- The Family Environment Scale had acceptable reliability (Cronbach's Alpha = 0.693). This means the five items were fairly consistent in measuring family environment. The item-total correlations showed that all items were useful, and the Alpha-if-Deleted values showed that removing any item would not improve the scale much. Therefore, all items were kept for further analysis.
- Therefore, the composite Family Environment and Mental Well-Being scores were appropriate for use in the subsequent correlation, group-comparison, and regression analyses.

4.8 Normality Testing

Table 24: Normality test results

Variable	N	Shapiro-Wilk Stat	Shapiro-Wilk p-value	K-S Stat	K-S p-value	Is Normal ?
Mental Development Score	339	0.9398	1.68e-10	0.1272	3.06e-05	No
Family Environment Score	339	0.9508	3.23e-09	0.1046	0.0011	No
Child Age	339	0.8636	1.09e-16	0.1814	3.11e-10	No

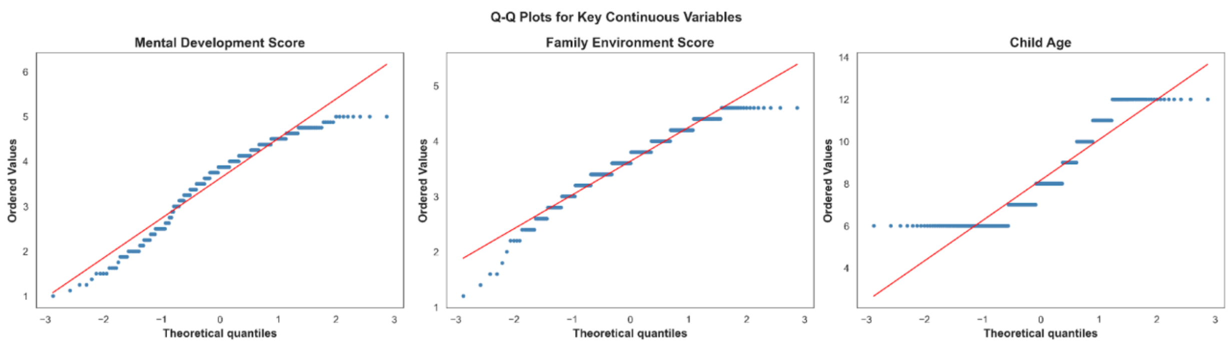


Figure 12: Q-Q plots for key variables

Interpretation

- Normality testing was conducted for three key continuous variables: Mental Development Score, Family Environment Score, and Child Age.
- Both the Shapiro-Wilk test and Kolmogorov-Smirnov test showed significant p-values for all three variables ($p < 0.05$), indicating that none of the variables followed a normal distribution.
- The Q-Q plots confirm departures from the diagonal reference line
- Because the normality assumption was not fully satisfied, the analysis appropriately used non-parametric tests such as Spearman correlation, Mann-Whitney U, and Kruskal-Wallis tests alongside parametric models where relevant. This strengthens the robustness of the findings.

4.9 Bivariate Analysis

4.9.1 Correlation Analysis

Correlation analysis was conducted to examine the relationship between the family environment variables and children's mental well-being. The family environment variables included emotional atmosphere, argument frequency, emotional safety, parent-child quality time, and family learning support. The overall Family Environment Score, Mental Well-Being Score, and Child Age were also included in the analysis.

Since the main variables were not normally distributed, Spearman's correlation was used. This analysis helped identify which aspects of the family environment were more strongly associated with children's mental well-being.

Table 25: Spearman correlation matrix

Category	Emotional Atmosphere	Argument Frequency	Emotional Safety	Parent Child Quality Time	Family Learning Support	Family Env	Mental WB	Child Age
Emotional Atmosphere	1.000	0.465	0.422	0.284	0.316	0.777	0.287	-0.073NS
Argument Frequency	0.465	1.000	0.233	0.165	0.223	0.600	0.235	-0.125
Emotional Safety	0.422	0.233	1.000	0.311	0.353	0.725	0.410	-0.086
Parent Child Quality Time	0.284	0.165	0.311	1.000	0.296	0.523	0.454	-0.216
Family Learning Support	0.316	0.223	0.353	0.296	1.000	0.607	0.330	-0.142
Family Env	0.777	0.600	0.725	0.523	0.607	1.000	0.466	-0.175
Mental WB	0.287	0.235	0.410	0.454	0.330	0.466	1.000	-0.270
Child Age	-0.073	-0.125	-0.086	-0.216	-0.142	-0.175	-0.270	1.000

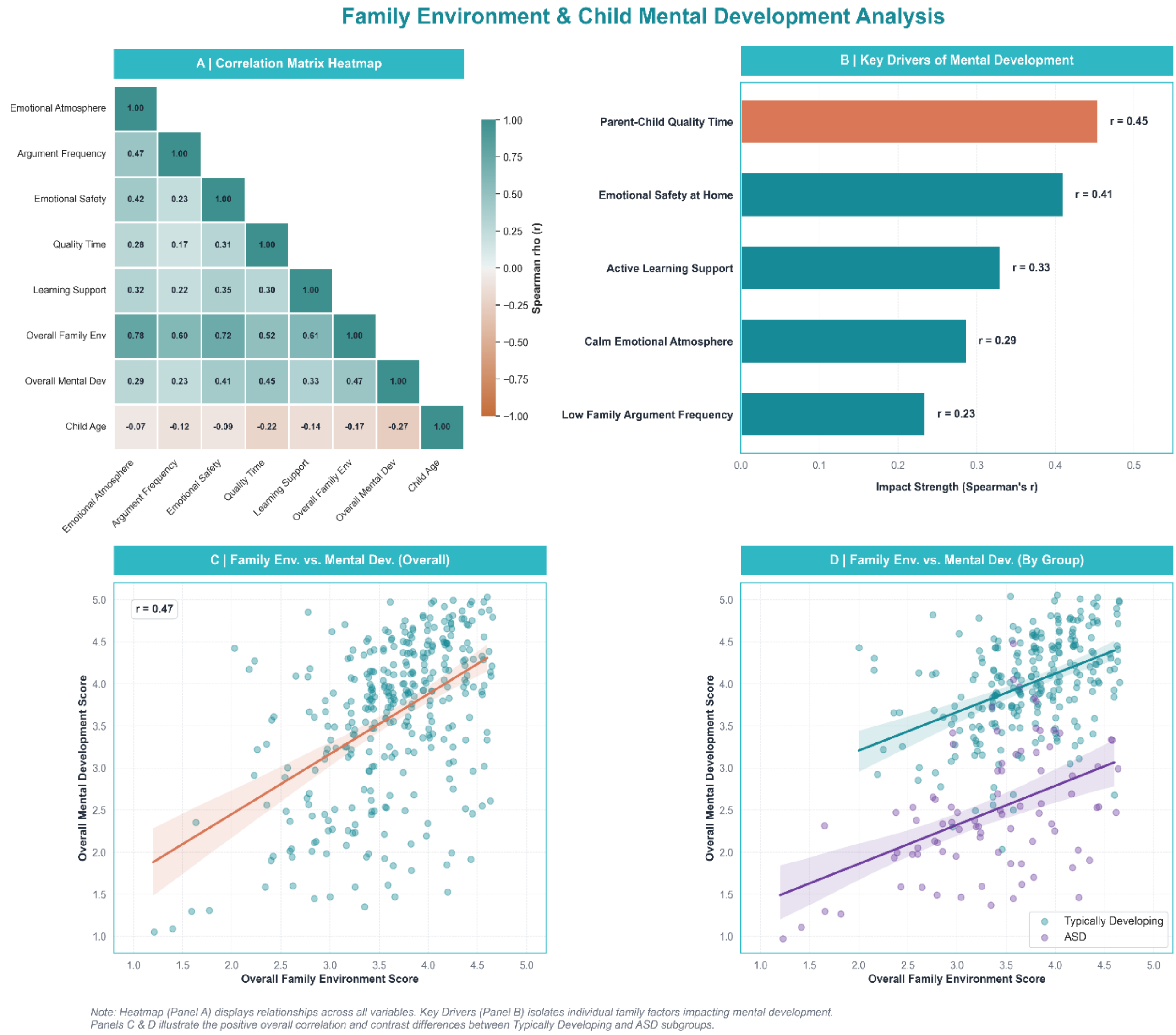


Figure 13: Correlation Heatmap, Scatter plots

This correlation analysis answers these main bivariate research questions:

Q1: Is family environment associated with mental well-being?

Yes. r =0.47 between Family_Env_Score × Mental_WB_Score

Q2: Which family environment indicators are most related to children's mental well-being?

Parent-child quality time r = 0.45

Q3: Do children with better family environments tend to have better mental well-being?

Yes. The scatter plot of Family_Env_Score × Mental_WB_Score showed an **upward trend**

Q4: Do TD children generally show higher mental well-being than ASD children in the scatter plot?

Yes. The grouped scatter plot suggests that TD children generally had **higher mental well-being scores** than ASD children overall.

Interpretation

Correlation Matrix Heatmap: The heatmap shows that most family environment indicators were positively related to mental well-being. The overall Family Environment Score had a moderate positive relationship with Mental Well-Being Score ($r = 0.47$).

Key Drivers of Mental Development: Parent-child quality time ($r = 0.45$) and emotional safety ($r = 0.41$) were the strongest family-related factors, followed by learning support ($r = 0.33$), calm emotional atmosphere ($r = 0.29$), and low argument frequency ($r = 0.23$).

Family Environment vs. Mental Well-Being: The scatter plot shows an upward trend. Children with higher Family Environment Scores generally had higher Mental Well-Being Scores.

Family Environment vs. Mental Well-Being by Group: Both TD and ASD children showed a positive relationship between family environment and mental well-being. However, TD children had higher mental well-being scores overall than children with ASD.

The correlation analysis suggests that children's mental well-being is more strongly linked with the quality of family care, emotional safety, quality time, and learning support than with family structure alone.

4.9.2 Chi-Square Tests of Independence

The Pearson Chi-Square test of independence was used to examine whether categorical variables are associated with Child Group (ASD vs. Typically Developing). The effect size is measured using Cramér's V.

Association between Family Structure and Child Group

Hypothesis:

H_0: Family structure and child group are independent.

H_1: Family structure and child group are associated.

Table 26: Cross Tabulation — Family Structure × Child Group

Family Structure	ASD	TD	Total
Joint Family	26 (30.6%)	82 (32.3%)	108 (31.9%)
Nuclear Family	53 (62.4%)	164 (64.6%)	217 (64.0%)
Single-Parent Family	6 (7.1%)	8 (3.1%)	14 (4.1%)
Total	85 (25.1%)	254 (74.9%)	339 (100.0%)

Table 27: Chi-Square Test — Family Structure × Child Group

Variables	χ^2	df	p-value	Cramér's V	Effect	Significant
Family_Type × Child_Group	2.463	2	0.292	0.085	Very weak	No

Comment: The result was not statistically significant, $\chi^2(2) = 2.46$, $p = 0.292$. There is no strong evidence that child group differs across family structures. The effect size was Cramér's V = 0.085, suggesting a very weak association. Therefore, H_0 is not rejected.

Association between Screen Time and Child Group

Hypothesis:

H_0: Screen time and child group are independent.

H_1: Screen time and child group are associated.

Table 28: Cross Tabulation — Screen Time × Child Group

Screen Time	ASD	TD	Total
1-2 hours	31 (36.5%)	103 (40.6%)	134 (39.5%)
3-4 hours	23 (27.1%)	48 (18.9%)	71 (20.9%)
Less than 1 hour	13 (15.3%)	74 (29.1%)	87 (25.7%)
More than 4 hours	17 (20.0%)	16 (6.3%)	33 (9.7%)
No Screen Use	1 (1.2%)	13 (5.1%)	14 (4.1%)
Total	85 (25.1%)	254 (74.9%)	339 (100.0%)

Table 29: Chi-Square Test — Screen Time × Child Group

Variables	χ^2	df	p-value	Cramér's V	Effect	Significant
Screen_Time × Child_Group	21.724	4	< .001	0.253	Weak	Yes

Comment: The result was statistically significant, $\chi^2(4) = 21.72$, $p < .001$. ASD children show higher rates of heavy screen use: 20% use screens 4+ hours/day vs. only 6.3% of TD children. Cramér's V = 0.253 (weak association). H_0 is rejected.

Association between Screen Content and Child Group

Hypothesis:

H_0: Screen content and child group are independent.

H_1: Screen content and child group are associated.

Table 30: Chi-Square Test — Screen Content × Child Group

Variables	χ^2	df	p-value	Cramér's V	Effect	Significant
Screen_Content × Child_Group	8.074	5	0.152	0.154	Weak	No

Comment: The result was not statistically significant, $\chi^2(5) = 8.07$, $p = 0.152$. There is no strong evidence that screen content type differs between TD and ASD children. H_0 is not rejected.

Association between Screen Use Time and Child Group

Hypothesis:

H_0: Screen use time (time of day) and child group are independent.

H_1: Screen use time and child group are associated.

Table 31: Cross Tabulation — Screen Use Time × Child Group

Time of Day	ASD	TD	Total
Afternoon	24 (28.2%)	45 (17.7%)	69 (20.4%)
Evening	34 (40.0%)	48 (18.9%)	82 (24.2%)
Morning	7 (8.2%)	8 (3.1%)	15 (4.4%)
Night	16 (18.8%)	70 (27.6%)	86 (25.4%)
Noon	3 (3.5%)	70 (27.6%)	73 (21.5%)
Total	85 (25.1%)	254 (74.9%)	339 (100.0%)

Table 32: Chi-Square Test — Screen Use Time × Child Group

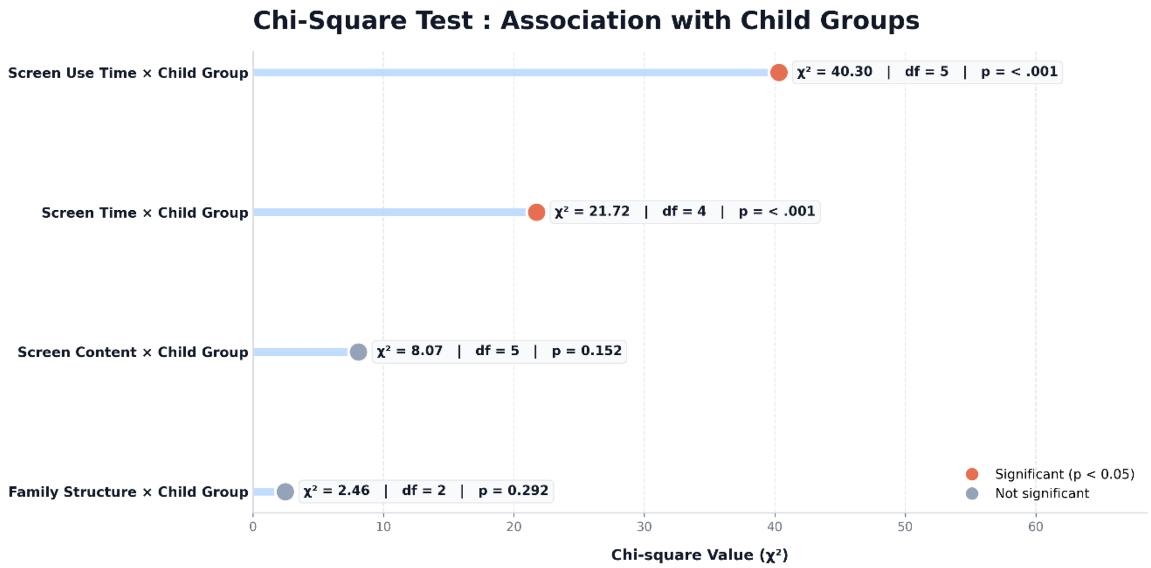
Variables	χ^2	df	p-value	Cramér's V	Effect	Significant
Screen_Use_Time × Child_Group	40.299	5	< .001	0.345	Moderate	Yes

Comment: The result was statistically significant, $\chi^2(5) = 40.30$, $p < .001$. This is the strongest chi-square association found. ASD children predominantly use screens in the evening (40.0%) and afternoon (28.2%), while TD children prefer noon (27.6%) and night (27.6%). H_0 is rejected.

Summary of Chi-Square Tests

Table 33: Summary of All Chi-Square Test Results

Variables	χ^2	df	p-value	Cramér's V	Effect	Significant
Family_Type × Child_Group	2.463	2	0.292	0.085	Very weak	No
Screen_Time × Child_Group	21.724	4	< .001	0.253	Weak	Yes
Screen_Content × Child_Group	8.074	5	0.152	0.154	Weak	No
Screen_Use_Time × Child_Group	40.299	5	< .001	0.345	Moderate	Yes



4.9.3 Mann-Whitney U Tests

The Mann-Whitney U test was used to compare continuous scores between two independent groups. The effect size is measured using the rank-biserial correlation (r).

Mental Well-Being: TD vs. ASD Children

Hypothesis:

H_0: There is no difference in mental well-being scores between TD and ASD children.

H_1: Mental well-being scores differ significantly between the two groups.

Table 34: Descriptive Statistics — Mental Well-Being by Child Group

Child Group	N	Mean	SD	Median	Min	Max
Typically Developing	254	3.998	0.582	4.000	2.0	5.0
ASD	85	2.482	0.740	2.375	1.0	4.5

Table 35: Mann-Whitney U Test — Mental Well-Being by Child Group

Comparison	Score	U	p-value	r	Effect	Significant
TD vs. ASD	Mental WB Score	20,241.0	< .001	−0.875	Strong	Yes

Comment: The result was statistically significant, U = 20,241.00, $p < .001$. TD children scored significantly higher (M = 4.00) than ASD children (M = 2.48). The rank-biserial correlation r = −0.875 suggests a strong effect. This is the largest effect observed across all analyses. H_0 is rejected.

Mental Well-Being: Male vs. Female Children

Hypothesis:

H_0: There is no difference in mental well-being scores between male and female children.

H_1: Mental well-being scores differ significantly by gender.

Table 36: Mann-Whitney U Test — Mental Well-Being by Gender

Comparison	Score	U	p-value	r	Effect	Significant
Male vs. Female	Mental WB Score	12,535.0	0.388	0.056	Very weak	No

Comment: The result was not statistically significant, U = 12,535.00, p = 0.388. Males (M = 3.58) and females (M = 3.68) had very similar scores. Gender is not a meaningful predictor of mental well-being. H_0 is not rejected.

Family Environment: TD vs. ASD Children

Hypothesis:

H_0: There is no difference in family environment scores between TD and ASD children.

H_1: Family environment scores differ significantly between the two groups.

Table 37: Mann-Whitney U Test — Family Environment by Child Group

Comparison	Score	U	p-value	r	Effect	Significant
TD vs. ASD	Family Env Score	14,233.5	< .001	−0.319	Moderate	Yes

Comment: The result was statistically significant, U = 14,233.50, p < .001. TD families scored higher (M = 3.73) than ASD families (M = 3.34), suggesting less favorable home environments in ASD families. The rank-biserial r = −0.319 indicates a moderate effect. H_0 is rejected.

Summary of Mann-Whitney U Tests

Table 38: Summary of All Mann-Whitney U Test Results

Comparison	Score	U	p-value	r	Effect	Significant
TD vs. ASD	Mental WB	20,241.0	< .001	−0.875	Strong	Yes
Male vs. Female	Mental WB	12,535.0	0.388	0.056	Very weak	No
TD vs. ASD	Family Env	14,233.5	< .001	−0.319	Moderate	Yes

4.9.4 Kruskal-Wallis Tests

The Kruskal-Wallis test was used to compare continuous scores across three or more independent groups. The effect size is measured using epsilon-squared (ε^2).

Mental Well-Being across Family Structures

Hypothesis:

H_0: Mental well-being scores do not differ across family structures.

H_1: Mental well-being scores differ significantly across family structures.

Table 39: Kruskal-Wallis Test — Mental Well-Being across Family Structures

Grouping Var	Score Var	H	df	p-value	ε²	Effect	Sig?
Family_Type	Mental WB	3.245	2	0.197	0.004	Very weak	No

Comment: The result was not statistically significant, $H(2) = 3.25$, $p = 0.197$. Mental well-being does not differ across family structures. H_0 is not rejected.

Family Environment across Family Structures

Hypothesis:

H_0: Family environment scores do not differ across family structures.

H_1: Family environment scores differ significantly across family structures.

Table 40: Kruskal-Wallis Test — Family Environment across Family Structures

Grouping Var	Score Var	H	df	p-value	ε²	Effect	Sig?
Family_Type	Family Env	11.640	2	0.003	0.029	Weak	Yes

Comment: The result was statistically significant, $H(2) = 11.64$, $p = 0.003$. Nuclear families report the highest scores, single-parent families the lowest. H_0 is rejected.

Mental Well-Being by Mother's Education

Hypothesis:

H_0: Mental well-being scores do not differ across mother's education levels.

H_1: Mental well-being scores differ significantly across mother's education levels.

Table 41: Kruskal-Wallis Test — Mental Well-Being by Mother's Education

Grouping Var	Score Var	H	df	p-value	ε²	Effect	Sig?
Mother_Education	Mental WB	45.888	7	< .001	0.117	Moderate	Yes

Comment: The result was statistically significant, $H(7) = 45.89$, $p < .001$. This is the strongest Kruskal-Wallis effect ($\varepsilon^2 = 0.117$, moderate). Children of mothers with Master's degrees had highest well-being; primary education had the lowest. H_0 is rejected.

Mental Well-Being by Father's Education

Hypothesis:

H_0: Mental well-being scores do not differ across father's education levels.

H_1: Mental well-being scores differ significantly across father's education levels.

Table 42: Kruskal-Wallis Test — Mental Well-Being by Father's Education

Grouping Var	Score Var	H	df	p-value	ε²	Effect	Sig?
Father_Education	Mental WB	20.463	8	0.009	0.038	Weak	Yes

Comment: The result was statistically significant, $H(8) = 20.46$, $p = 0.009$. H_0 is rejected.

Family Environment by Father's Occupation

Hypothesis:

H_0: Family environment scores do not differ across father's occupation types.

H_1: Family environment scores differ significantly across father's occupation types.

Table 43: Kruskal-Wallis Test — Family Environment by Father's Occupation

Grouping Var	Score Var	H	df	p-value	ε²	Effect	Sig?
Father_Occupation	Family Env	8.413	7	0.297	0.004	Very Weak	No

Comment: The result was not statistically significant, $H(7) = 8.41$, $p = 0.297$. Therefore, family environment scores did not differ significantly across father's occupation types.

Family Environment by Mother's Occupation

Hypothesis:

H_0: Family environment scores do not differ across mother's occupation types.

H_1: Family environment scores differ significantly across mother's occupation types.

Table 44: Kruskal-Wallis Test — Family Environment by Mother's Occupation

Grouping Var	Score Var	H	df	p-value	ε²	Effect	Sig?
Mother_Occupation	Family Env	11.558	4	0.021	0.023	Weak	Yes

Comment: The result was statistically significant, $H(4) = 11.56$, $p = 0.021$. Homemaker mothers had the highest family environment scores. H_0 is rejected.

Mental Well-Being across Screen Time

Hypothesis:

H_0: Mental well-being scores do not differ across screen time categories.

H_1: Mental well-being scores differ significantly across screen time categories.

Table 45: Kruskal-Wallis Test — Mental Well-Being across Screen Time Categories

Grouping Var	Score Var	H	df	p-value	ε^2	Effect	Sig?
Screen_Time	Mental WB	13.928	3	0.003	0.033	Weak	Yes

Comment: The result was statistically significant, $H(3) = 13.93$, $p = 0.003$. Children with no screen use scored highest (Mdn = 4.00), while 3–4 hours group scored lowest (Mdn = 3.63). H_0 is rejected.

Summary of All Kruskal-Wallis Tests

Table 46: Complete Kruskal-Wallis Test Summary

Grouping Var	Score Var	H	df	p-value	ε^2	Effect	Sig?
Family_Type	Mental WB	3.245	2	0.197	0.004	Very weak	No
Family_Type	Family Env	11.640	2	0.003	0.029	Weak	Yes
Mother_Education	Mental WB	45.888	7	< .001	0.117	Moderate	Yes
Father_Education	Mental WB	20.463	8	0.009	0.038	Weak	Yes
Father_Occupation	Family Env	8.413	7	0.296	0.004	Very Weak	No
Mother_Occupation	Family Env	11.558	4	0.021	0.023	Weak	Yes
Screen_Time	Mental WB	13.928	3	0.003	0.033	Weak	Yes

4.10 Multivariate Analysis

4.10.1 Variable Selection for Multivariate Modeling

Variables for the multivariate model were selected based on the study objectives, theoretical relevance, bivariate findings, and model stability. Mental Well-Being Score was used as the dependent variable. Family Environment Score, Family Type, and Child Group were retained as key predictors because they were directly related to the study objectives. Child age, gender, household size, screen time, parental education, and monthly income were included as control or background variables. Variables with sparse categories, weak relevance, or strong overlap with other predictors were excluded to improve model stability and interpretation.

Basis of Variable Selection

Table 47: Variable selection basis for multivariate modeling

Selection Basis	Meaning in This Study	Variables Included
Research objective	Directly related to the main aim of the study.	Mental Well-Being Score, Family Environment Score, Family Type
Theoretical relevance	Variables that may influence children's mental well-being were included	Child Age, Child Gender, Household Size
Bivariate evidence	Showed important findings in earlier bivariate analysis.	Screen Time, Father's Education, Mother's Education
Socioeconomic control	Represents family background and economic condition.	Father's Education, Mother's Education, Monthly Income
Study design / group comparison	Needed because the study compares ASD and TD children.	Child Group
Model stability	Variables with too many small categories or overlapping information were excluded.	Excluded: School Status, Primary Caregiver, Screen Content, Screen Use Time, Father's Occupation, Mother's Occupation

Table 48: Final model variable plan

Variable Group	Variables
Outcome Variable	Mental Well-Being Score
Selected Predictor Variables	Family Environment Score, Family Type, Child Group, Child Age, Child Gender, Household Size, Screen Time, Father's Education, Mother's Education, Monthly Income
Excluded Variables	Primary Caregiver, School Status, Screen Content, Screen Use Time, Father's Occupation, Mother's Occupation, Family Environment Item Scores, Mental Well-Being Item Scores

Final Model Plan

Table 49: Hierarchical regression model block plan

Model	Variables Added	Purpose
Model 1	Child Age, Child Gender, Child Group	Controls basic child characteristics and ASD/TD group difference.
Model 2	Model 1 + Family Type, Household Size	Tests whether family structure and household context add explanatory value.
Model 3	Model 2 + Family Environment Score	Tests whether family environment explains mental well-being beyond child and family structure factors.
Model 4	Model 3 + Screen Time	Examines whether screen exposure adds additional explanatory value.
Model 5	Model 4 + Father's Education, Mother's Education, Monthly Income	Final adjusted model controlling for socioeconomic background.

4.10.2 Multicollinearity Check

Before fitting the regression model, multicollinearity among the selected predictors was checked using the Variance Inflation Factor (VIF). A total of 32 predictors were examined using 339 observations.

The highest VIF value was 2.764, which was below the commonly used cutoff value of 5. All predictors had VIF values below 5, and no predictor pair showed a correlation of 0.80 or higher. Therefore, no serious multicollinearity problem was found.

Table 50: Multicollinearity check results (VIF summary)

Item	Result
Observations used	339
Predictors checked	32
Maximum VIF	2.764
Predictors with VIF < 5	32
Predictors with VIF ≥ 5	0
High pairwise correlation ≥ 0.80	None

Interpretation:
The selected predictors were suitable for regression analysis. No variable needed to be removed due to multicollinearity.

4.10.3 Multiple Linear Regression Analysis

Multiple linear regression was conducted to identify the adjusted predictors of children's Mental Well-Being Score. The model included child demographic factors, family structure, Family Environment Score, screen time, parental education, and monthly income as predictors.

The overall regression model was statistically significant, $F(32, 306) = 20.753$, $p < .001$, indicating that the selected predictors collectively explained a significant amount of variation in children's Mental Well-Being Score. The model explained 68.5% of the variance in Mental Well-Being Score, with an adjusted R^2 of 65.2%. This suggests that the model had strong explanatory power.

Summary of Model Fit

Table 51: Multiple linear regression model fit summary

Model Fit Indicator	Value
N	339
Number of predictors	32
R^2	0.685
Adjusted R^2	0.652
F-statistic	20.753
Model p-value	< .001
AIC	569.496
BIC	695.754
RMSE	0.5085
MAE	0.3997

Significant Adjusted Predictors

Table 52: Significant adjusted predictors of mental well-being

Predictor	B	95% CI	p-value	Interpretation
Family Environment Score	0.414	0.309 to 0.519	< .001	Better family environment was associated with higher mental well-being.
Child Group: ASD	-1.204	-1.358 to -1.050	< .001	ASD children had lower mental well-being than typically developing children.
Mother Education: HSC/12	-0.271	-0.481 to -0.061	0.012	Lower than reference group.
Mother Education: No formal schooling	-0.623	-1.063 to -0.183	0.006	Lower than reference group.

Mother Education: Primary	-0.675	-1.007 to -0.342	< .001	Lower than reference group.
Mother Education: SSC/Class 10	-0.295	-0.527 to -0.064	0.013	Lower than reference group.

The reference group for Child Group was Typically Developing children, and the reference group for Mother Education was Master's Degree or Higher.

Interpretation

After adjusting for child age, gender, household size, family type, screen time, father's education, mother's education, and monthly income, Family Environment Score remained a significant positive predictor of children's Mental Well-Being Score. Specifically, a one-unit increase in Family Environment Score was associated with a 0.414-unit increase in Mental Well-Being Score, holding other variables constant.

Child group was also a strong predictor. Children with ASD had significantly lower Mental Well-Being Scores than typically developing children, with an adjusted coefficient of $B = -1.204$, $p < .001$. This indicates that, after controlling for other factors, ASD status was associated with lower mental well-being.

Mother's education was another important predictor. Compared with children whose mothers had a Master's degree or higher, children whose mothers had Higher Secondary education, Secondary education, Primary education, or no formal schooling had significantly lower Mental Well-Being Scores. Among these, the strongest negative coefficient was observed for Primary education.

Other variables, including child age, child gender, household size, family type, screen time, father's education, and monthly income, were not statistically significant in the adjusted model.

Overall, the multiple linear regression results showed that family environment, child group, and mother's education were significant adjusted predictors of children's mental well-being. Better family environment was associated with higher mental well-being, while ASD status and lower maternal education levels were associated with lower mental well-being.

4.10.4 Hierarchical Regression Analysis

Hierarchical regression was conducted to examine how different groups of predictors contributed to children's Mental Well-Being Score. The predictors were entered step by step in five blocks: child factors, family structure and household context, family environment, screen exposure, and socioeconomic background.

The first model included only child age, child gender, and child group. This model explained 54.3% of the variation in Mental Well-Being Score, showing that child-level factors, especially child group, were important in explaining children's mental well-being.

After adding family type and household size in Model 2, the explained variance increased only slightly from 54.3% to 54.5%. This change was not statistically significant, $\Delta R^2 = 0.0027$, $p = 0.5848$. This means that family structure and household size did not add meaningful explanatory power after controlling for child factors.

The most important improvement occurred in Model 3, when Family Environment Score was added. The explained variance increased from 54.5% to 62.9%, and this improvement was statistically significant, $\Delta R^2 = 0.0839$, $p < .001$. This indicates that family environment made a strong additional contribution to children's mental well-being beyond child factors and family structure.

In Model 4, screen time was added and produced a small but significant improvement, $\Delta R^2 = 0.0164$, $p = 0.0051$. Finally, Model 5 added father's education, mother's education, and monthly income. The final model explained 68.5% of the variance in Mental Well-Being Score, with an adjusted R^2 of 65.2%. The socioeconomic block also significantly improved the model, $\Delta R^2 = 0.0391$, $p = 0.0175$.

Hierarchical Model Comparison

Table 53: Hierarchical regression model comparison

Model	Block Added	R^2	Adjusted R^2	ΔR^2	p-value for Change	Decision
Model 1	Child factors	0.5425	0.5384	Baseline	Baseline	Baseline model
Model 2	Family structure and household context	0.5452	0.5370	0.0027	0.5848	No significant improvement
Model 3	Family environment	0.6291	0.6212	0.0839	< .001	Significant improvement
Model 4	Screen exposure	0.6455	0.6335	0.0164	0.0051	Significant improvement
Model 5	Socioeconomic background	0.6846	0.6516	0.0391	0.0175	Significant improvement

Final Model Interpretation

In the final hierarchical model, Family Environment Score, Child Group, and selected categories of Mother's Education were statistically significant predictors of Mental Well-Being Score. Family Environment Score showed a positive association, $B = 0.4143$, $p < .001$, meaning that children from better family environments had higher mental well-being scores after adjustment. Child Group was also a strong predictor; children with ASD had significantly lower Mental Well-Being Scores than typically developing children, $B = -1.2040$, $p < .001$.

Mother's education remained important in the final model. Compared with mothers having a Master's degree or higher, children whose mothers had Higher Secondary education, Secondary education, Primary education, or no formal schooling had significantly lower Mental Well-Being Scores. Other variables, including child age, gender, household size, family type, screen time categories, father's education, and monthly income, were not significant in the final adjusted model.

Overall, the hierarchical regression analysis showed that family environment added substantial explanatory power to the model after controlling for child factors and family structure. The final model explained 68.5% of the variance in children's Mental Well-Being Score, with Family Environment Score, Child Group, and Mother's Education emerging as the main adjusted predictors.

4.10.5 Regression Diagnostics

Regression diagnostics were performed to assess whether the final linear regression model was suitable for interpretation. The residual normality assumption was acceptable, as both the Shapiro-Wilk test, $p = 0.0627$, and Jarque-Bera test, $p = 0.0604$, were not significant. The Breusch-Pagan test was also not significant, $p = 0.2424$, indicating no serious heteroscedasticity problem. The Durbin-Watson statistic was 1.8155, suggesting acceptable independence of residuals.

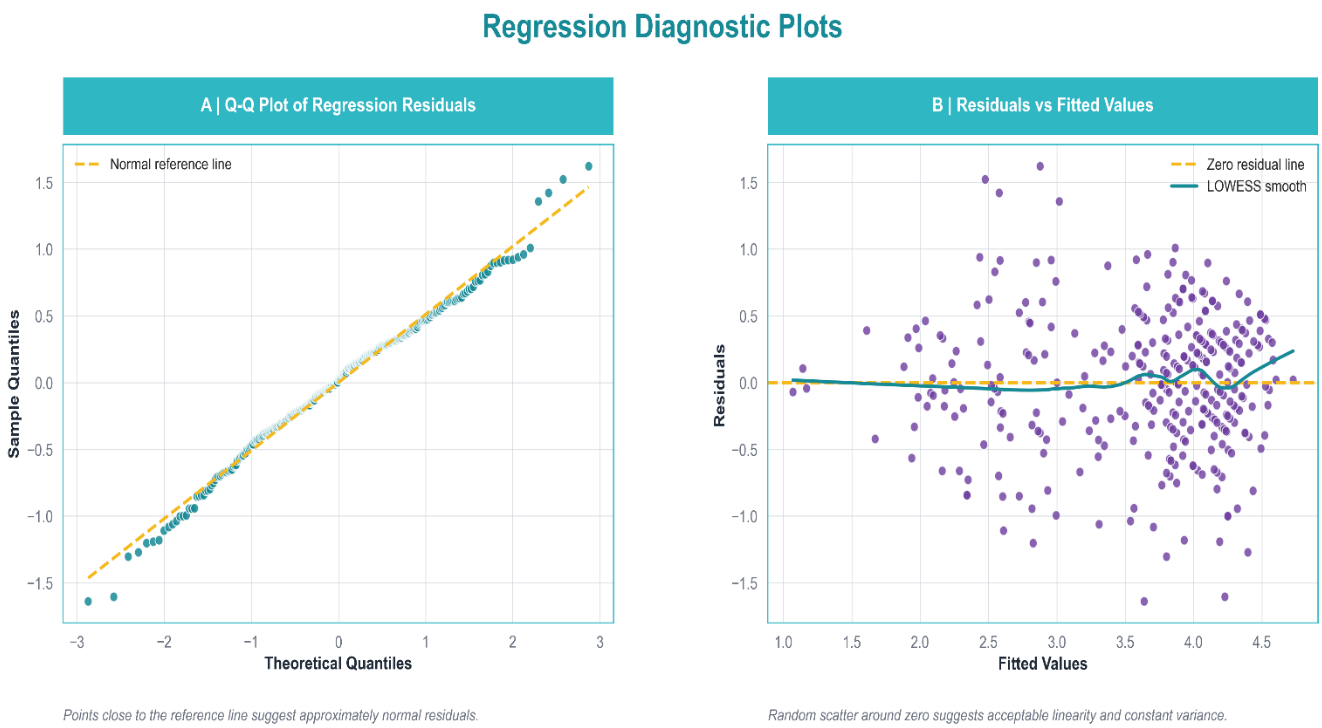


Table 54: Regression diagnostics summary

Diagnostic Check	Result	Acceptable Rule	Decision
Residual Normality	Shapiro-Wilk p = 0.0627; Jarque-Bera p = 0.0604	p-value ≥ 0.05 suggests residuals do not significantly deviate from normality	Acceptable
Equal Variance / Homoscedasticity	Breusch-Pagan p = 0.2424	p-value ≥ 0.05 suggests no serious heteroscedasticity	Acceptable
Residual Independence	Durbin-Watson = 1.8155	Values close to 2, commonly 1.5–2.5, are acceptable	Acceptable
Influential Cases	26 cases above Cook's distance threshold	Cases above the selected Cook's distance cutoff should be reviewed	Review Cautiously

Overall, the main regression assumptions were reasonably satisfied. Some influential observations were detected, but they were not removed automatically because influential cases should only be excluded if they are clear data-entry errors or invalid responses. Therefore, the final regression model was considered acceptable for interpretation.

4.10.6 Moderation / Interaction Analysis

Interaction analysis was conducted to examine whether the effect of Family Environment Score on Mental Well-Being Score changed across child group and family type.

Table 55: Moderation / interaction research questions

Interaction Tested	Research Question
Family Environment × Child Group	Does the relationship between family environment and mental well-being differ between ASD and typically developing children?
Family Environment × Family Type	Does the relationship between family environment and mental well-being differ across nuclear, joint, and single-parent families?

Table 56: Interaction model fit comparison

Model	N	R^2	Adjusted R^2	F-statistic	Model p-value	AIC	BIC
Base Main-Effects Model	339	0.6846	0.6516	20.753	< .001	569.496	695.754
Family Environment × Child Group	339	0.6846	0.6505	20.060	< .001	571.476	701.560
Family Environment × Family Type	339	0.6938	0.6595	20.258	< .001	563.442	697.352

Table 57: Interaction significance test results

Interaction Tested	Base Model R^2	Interaction Model R^2	ΔR^2	F Change	df Change	p-value	Decision
Family Environment × Child Group	0.6846	0.6846	0.0000	0.018	1	0.892	No significant improvement
Family Environment × Family Type	0.6846	0.6938	0.0092	4.576	2	0.011	Significant improvement

Table 58: Family environment effect by family type and child group

Interaction Model	ΔR^2	p-value	Decision	Interpretation
Family Environment × Child Group	0.0000	0.892	Not significant	The effect of family environment did not differ between ASD and TD children.
Family Environment × Family Type	0.0092	0.011	Significant	The effect of family environment differed across family structures.

The interaction between Family Environment Score and Child Group was not statistically significant, $\Delta R^2 = 0.0000$, $p = 0.892$. This indicates that the relationship between family environment and mental well-being did not differ significantly between ASD and typically developing children. In both groups, better family environment was significantly associated with higher mental well-being.

However, the interaction between Family Environment Score and Family Type was statistically significant, $\Delta R^2 = 0.0092$, $p = 0.011$. This suggests that the effect of family environment on mental well-being differed across family structures. The positive effect of family environment was strongest among children from joint families, followed by nuclear families. For single-parent families, the relationship was positive but not statistically significant, which should be interpreted cautiously because the single-parent group was s mall.

Family Environment Effect by Child Group

Table 59: Family environment effect by child group

Child Group	Slope of Family Environment Score	SE	t	p-value	Decision	Interpretation
Typically Developing	0.4198	0.0674	6.2305	< .001	Significant	Better family environment was associated with higher mental well-being.
ASD	0.4053	0.0846	4.7904	< .001	Significant	Better family environment was also associated with higher mental well-being among ASD children.

Family Environment Score had a significant positive effect on Mental Well-Being Score for both typically developing children and ASD children. However, because the interaction between Family Environment and Child Group was not significant, the strength of this relationship did not differ meaningfully between the two child groups.

Family Environment Effect by Family Type

Table 60: Binary logistic regression hypothesis

Family Type	Slope of Family Environment Score	SE	t	p-value	Decision	Interpretation
Nuclear Family	0.3284	0.0651	5.0467	< .001	Significant	Better family environment was associated with higher mental well-being.
Joint Family	0.6200	0.0884	7.0141	< .001	Significant	The positive effect of family environment was strongest in joint families.
Single-Parent Family	0.1425	0.2085	0.6835	0.4948	Not significant	The relationship was positive but not statistically significant.

Family Environment Score was significantly associated with Mental Well-Being Score in nuclear and joint families. The effect was strongest among joint families (0.6200) then in nuclear family. In single-parent families, the effect was positive but not significant, possibly due to the small number of single-parent family cases.

4.10.7 Binary Logistic Regression

Binary logistic regression was used as an additional analysis to examine the factors related to high mental well-being among children. For this analysis, Mental Well-Being Score was converted into a binary outcome using the median cutoff value of 3.875. Children with scores equal to or above the median were classified as having high mental well-being, while those below the median were classified as having low mental well-being. The resulting outcome was nearly balanced, with 173 children (51.0%) in the high mental well-being group and 166 children (49.0%) in the low mental well-being group

What factors increase or decrease the odds of high mental well-being among children?

Table 61: Binary logistic regression model fit

Hypothesis Type	Statement
Null Hypothesis (H_0)	Family environment, child group, family type, screen time, parental education, and income are not significantly associated with the odds of high mental well-being.
Alternative Hypothesis (H_1)	At least one of these factors is significantly associated with the odds of high mental well-being.

Table 62: Significant predictors of high mental well-being (logistic regression)

Model	Outcome	N	Log-Likelihood	Null Log-Likelihood	McFadden Pseudo R^2	LLR p-value	AIC	BIC
Adjusted Binary Logistic Regression	High Mental Well-Being vs Low Mental Well-Being	339	-140.495	-234.905	0.4019	< .001	346.991	473.249

The logistic regression model was statistically significant, LLR $p < .001$, indicating that at least one of the factors is significantly associated with the odds of high mental well-being.

The McFadden pseudo R^2 value was 0.4019, suggesting good explanatory performance for a logistic regression model.

Significant Predictors of High Mental Well-Being

Table 63: Confusion matrix for binary logistic regression

Predictor	Adjusted OR	95% CI	p-value	Interpretation
Family Environment Score	5.8638	3.0461–11.2881	< .001	Better family environment increased the odds of high mental well-being.
Child Group: ASD	0.0219	0.0066–0.0722	< .001	ASD children had much lower odds of high mental well-being than TD children.
Screen Time: 1–2 hours	2.4013	1.0660–5.4089	0.0345	Children with 1–2 hours of screen time had higher odds of high mental well-being than the reference group.

Mother Education: Primary	0.0624	0.0087–0.4458	0.0057	Children whose mothers had primary education had lower odds of high mental well-being than those whose mothers had Master's degree or higher.

Table 64: Classification performance metrics

Actual Group	Predicted Low	Predicted High
Actual Low	127	39
Actual High	22	151

Table 65: Machine learning model comparison

Accuracy	Precision	Recall	F1-score	ROC-AUC
0.8201	0.7947	0.8728	0.8320	0.8931

The model showed good classification performance, with an accuracy of 82.0% and ROC-AUC of 0.8931. This indicates that the model was able to distinguish between low and high mental well-being groups reasonably well.

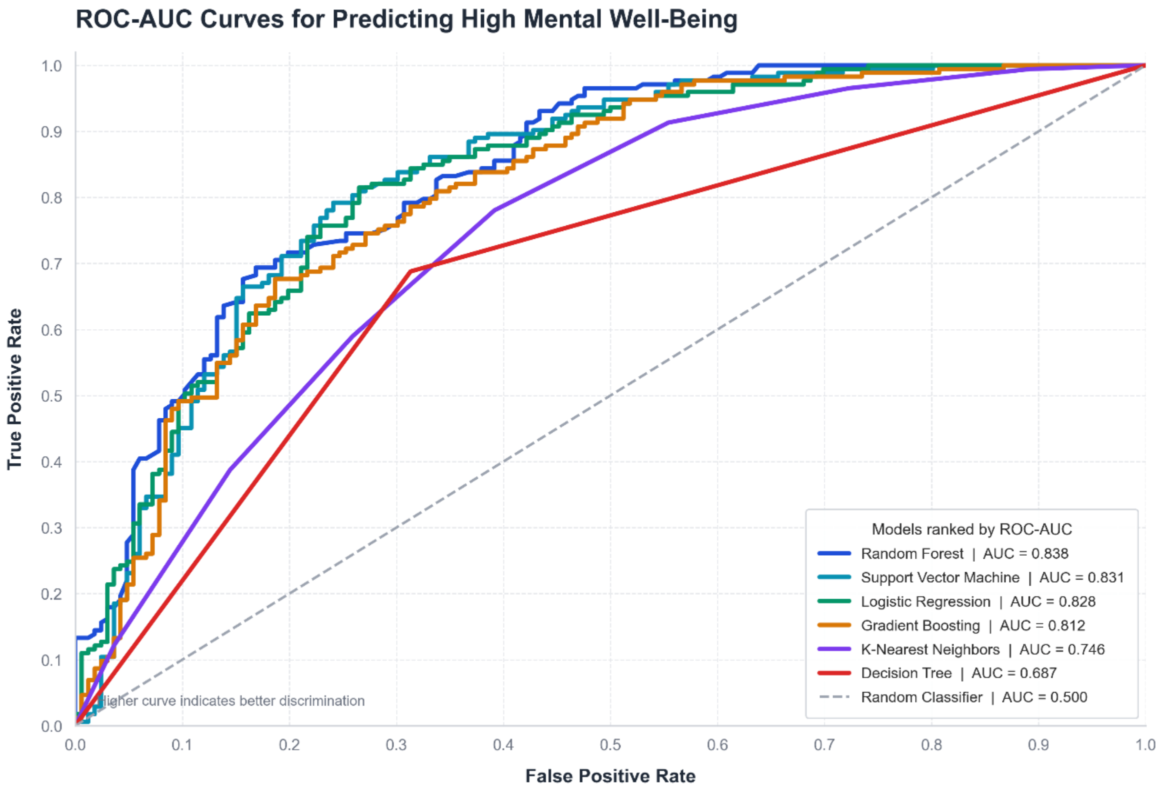
Overall, the supplementary logistic regression confirmed that better family environment increased the likelihood of high mental well-being, while ASD status and lower maternal education were associated with reduced odds of high mental well-being. The findings were consistent with the main linear regression results, although they should be interpreted cautiously due to convergence and sparse-category issues.

4.10.8 Machine Learning Model Comparison

A supplementary machine learning model comparison was conducted to evaluate how well different classification models could predict low vs high mental well-being. The binary outcome was created using the median Mental Well-Being Score cutoff of 3.875. Based on this cutoff, 173 children (51.0%) were classified as having high mental well-being and 166 children (49.0%) were classified as having low mental well-being. Six models were compared using 5-fold stratified cross-validation: Logistic Regression, Random Forest, Support Vector Machine, Gradient Boosting, K-Nearest Neighbors, and Decision Tree.

Table 66: ML Model Comparison

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Random Forest	0.7286	0.7263	0.7576	0.7402	0.8400
Logistic Regression	0.7727	0.7562	0.8207	0.7868	0.8324
Support Vector Machine	0.7699	0.7511	0.8271	0.7862	0.8296
Gradient Boosting	0.7285	0.7256	0.7573	0.7402	0.8146
K-Nearest Neighbors	0.6961	0.6766	0.7810	0.7233	0.7441
Decision Tree	0.6874	0.6997	0.6879	0.6925	0.6875



Interpretation

Among the six models, Random Forest achieved the highest ROC-AUC value of 0.8400, indicating the best overall ability to distinguish between low and high mental well-being groups. However, Logistic Regression showed the highest accuracy of 0.7727 and the highest F1-score of 0.7868, suggesting that it provided a strong balance between correct classification and prediction of the high mental well-being group. Support Vector Machine also performed similarly to Logistic Regression, with accuracy = 0.7699, F1-score = 0.7862, and ROC-AUC = 0.8296.

Overall, Random Forest performed best according to ROC-AUC, while Logistic Regression performed best according to accuracy and F1-score. Since Logistic Regression is easier to interpret and also performed strongly, it remains useful as the main supplementary classification model. The machine learning comparison was used only as an additional predictive analysis and does not replace the main regression-based explanatory findings.

Chapter 5: Discussion

5.1 Overview of Discussion

This study investigates the association of family environment and family structure with mental well-being of children with ASD and typically developing children in urban Pabna. The discussion highlights the key findings from descriptive, bivariate, regression, moderation and supplementary prediction analyses. The discussion centers on the principal findings from descriptive, bivariate, regression, moderation, and other prediction analyses.

5.2 Family Environment and Mental Well-Being

The study's main finding was that family environment was positively related to children's mental well-being. Children from family environments that were emotionally safe, calm, supportive and learning friendly scored better for mental wellbeing. This shows that the quality of family life is important for the emotional and behavioral development of children.

5.3 Family Structure vs Family Environment

Most of the children were from nuclear family followed by joint families and the smallest group was single parent families. Family structure was significant descriptively, but was not a strong predictor in the adjusted model. this indicates that the kind of family is less significant than the quality of the family environment. It means the emotional safety, quality time and family support matter, not the type of family whether it is nuclear, joint or single-parent.

5.4 ASD and Typically Developing Children

ASD children reported lower mental well-being than typically developing children. This may be because ASD children generally have more difficulties in communication, emotional regulation, social interaction, attention and adaptive behaviour. Therefore, ASD children may need more support from family, school and community. This may relate to the fact that ASD children tend to have more difficulties in communication, emotional regulation, social interaction, attention, and adaptive behavior. Thus, children with ASD may require more support from family, school, and community.

5.5 Role of Mother's Education

Mother's education remained important in the adjusted model. Children whose mothers had lower education levels showed lower mental well-being compared with children whose mothers had higher education. This may be because educated mothers may have better awareness of child development, emotional support, and caregiving practices. Father's education did not remain significant in the final adjusted model, so mother's education appeared stronger in this dataset.

5.6 Hierarchical Regression Findings

Hierarchical regression showed that child-level factors explained a large part of mental well-being. Family structure added little additional explanation, but Family Environment Score significantly improved the model. This confirms that family environment contributes to children's mental well-being beyond child group and family type.

5.7 Moderation Findings

Child group did not moderate the relationship between family environment and mental well-being. This means family environment was beneficial for both ASD and typically developing children. However, family type moderated the relationship. The effect of family environment was strongest in joint families, possibly because joint families may provide broader caregiving, emotional support, and practical help.

5.8 Supplementary Prediction Findings

The supplementary logistic regression and machine learning models supported the main findings. Family environment and child group were important for predicting high versus low mental well-being. However, these analyses were supplementary. The main interpretation of the study should be based on the multiple linear regression and hierarchical regression results.

5.9 Practical Implications

The findings suggest that parents and caregivers should focus on emotional safety, calm communication, parent-child quality time, and learning support. ASD children need additional structured support from families, schools, and communities. Parenting awareness programs may also be useful, especially for caregivers with lower educational backgrounds.

5.10 Strengths and Limitations

A strength of this study is that it included both ASD and typically developing children and used several levels of analysis, including descriptive, bivariate, multivariate, hierarchical, moderation, and supplementary prediction analysis. Reliability and diagnostic checks were also performed.

However, the study has some limitations. It was cross-sectional, so causal conclusions cannot be made. The study was conducted only in urban Pabna, so generalization may be limited. The single-parent family group was small, and some variables were self-reported. Logistic regression also had sparse-category and convergence concerns, so it was treated as supplementary.

5.11 Conclusion of Discussion

Overall, the study suggests that children's mental well-being is shaped more by the quality of the family environment than by family structure alone. Better emotional safety, calm family atmosphere, parent-child quality time, and learning support were linked with better mental well-being. ASD children showed lower mental well-being and may need additional support. Mother's education also appeared to be an important background factor. Therefore, interventions should focus on improving the emotional and supportive quality of the home environment, especially for children with ASD.

Chapter 6: Conclusion and Recommendations

6.1 Conclusion

In this study we explored the association of family environment and family structure with children's mental well-being among ASD and typically developing children in urban Pabna. Our findings indicate that family environment was more crucial than family structure alone.

Children who are from emotionally safe, calm, supportive, and learning-friendly families showed better mental well-being. Family Environment Score remained a significant positive predictor after adjusting for child group, family type, screen time, parental education, and income. ASD children had lower mental well-being than typically developing children, indicating a need for stronger family, school, and community support.

Only Family structure was not a strong direct predictor, but it moderated the relationship between family environment and mental well-being, with the strongest effect observed in joint families. Mother's education also remained important, suggesting that parenting awareness and emotional support practices influence children's well-being.

Overall, Our study concludes that children's mental well-being is shaped mainly by the quality of family care, emotional safety, and support, rather than family structure alone.

6.2 Major Conclusions of the Study

1. Family environment was positively associated with children's mental well-being. Children from emotionally safe and supportive families had better mental well-being.
2. ASD children had lower mental well-being than typically developing children. This highlights the need for extra family, school, and community support for ASD children.
3. Family structure alone was not a strong predictor of mental well-being. The quality of family relationships and support was more important than the type of family.

4. Mother's education was an important adjusted predictor.
   Higher maternal education may support better parenting awareness and child development practices.
5. Family type moderated the relationship between family environment and mental well-being.
   The effect of family environment was strongest in joint families.
6. Supplementary prediction models supported the main findings.
   Family environment and child group remained important in predicting children's mental well-being.

6.3 Recommendations

1. **Improve Family Environment**

   Parents and caregivers should focus on creating a supportive home environment through emotional safety, calm communication, quality time, and learning support.
2. **Provide Extra Support for ASD Children**

   ASD children should receive more structured support from family, school, and community, especially for communication, emotional regulation, social interaction, and adaptive behavior.
3. **Strengthen Parenting Awareness**

   Parenting education programs should be arranged to help caregivers understand children's emotional and developmental needs. These programs may focus on positive communication, emotional support, behavior management, and learning support at home.
4. **Support Mothers and Caregivers**

   Since mother's education was important, mothers and caregivers with lower educational background may benefit from practical guidance, awareness sessions, and child development training.
5. **Focus on Quality of Care, Not Only Family Type**

   Interventions should not focus only on whether a child lives in a nuclear, joint, or single-parent family. The main focus should be on the quality of support, emotional bonding, and caregiving inside the family.
6. **Promote Healthy Screen Habits**

   Parents should guide children toward balanced screen use by limiting excessive screen time, avoiding late-night screen use, and encouraging educational content, physical activity, and family interaction.
7. **Build School and Community Support**

Schools, special education centers, health workers, and community organizations should work with families to support children's mental well-being, especially for ASD children.

6.4 Recommendations for Future Research

Future studies may improve and extend this research in the following ways:

1. Use a larger and more balanced sample :Future studies should include more children from each family structure, especially single-parent families, to make comparisons more reliable.
2. Include both urban and rural areas : Future research may compare urban and rural settings to better understand how location affects family environment and children's mental well-being.
3. Study ASD children in more detail : Since ASD children showed lower mental well-being, future studies should explore their specific emotional, behavioral, and family support needs.
4. Use longitudinal design : Future research may follow children over time to understand how family environment influences mental well-being in the long term.
5. Use multiple sources of data : Teacher reports, clinical assessments, or psychologist-rated measures can be added with parent-reported data for stronger findings.
6. Include more family-related factors :Future studies may examine parenting style, caregiver stress, parental mental health, family communication, and social support.
7. Explore intervention programs :Future research can test whether parenting awareness, caregiver training, family counseling, or ASD support programs improve children's mental well-being.

References

Amato, P. R. (2005). The impact of family formation change on the cognitive, social, and emotional well-being of the next generation.

Bradley, R. H., & Corwyn, R. F. (2002). Socioeconomic status and child development.

Bronfenbrenner, U. (1979). *The ecology of human development*. Harvard University Press.

Cummings, E. M., & Davies, P. T. (2010). *Marital conflict and children: An emotional security perspective*. Guilford Press.

Demo, D. H., & Cox, M. J. (2000). Families with young children: A review of research.

Fleiss, J. L. (1981). *Statistical methods for rates and proportions*. John Wiley & Sons.

Kelsey, J. L., Whittemore, A. S., Evans, A. S., & Thompson, W. D. (1996). *Methods in observational epidemiology* (2nd ed.). Oxford University Press.

National Institute of Mental Health. (2019). *National Mental Health Survey of Bangladesh, 2018–2019*. Dhaka, Bangladesh.

OpenEpi. (n.d.). *Sample size for cross-sectional, cohort, and randomized clinical trial studies: OpenEpi version 3*.

Sharma, S. K., Mudgal, S. K., Thakur, K., & Gaur, R. (2020). How to calculate sample size for observational and experimental nursing research studies? *National Journal of Physiology, Pharmacy and Pharmacology, 10*(1), 1–8.

World Health Organization. (2022). Mental health and child development resources.

American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders* (5th ed.). American Psychiatric Publishing.

Baron-Cohen, S. (2008). *Autism and Asperger syndrome*. Oxford University Press.

Baumrind, D. (1991). The influence of parenting style on adolescent competence and substance use. *The Journal of Early Adolescence, 11*(1), 56–95.

Belsky, J. (1984). The determinants of parenting: A process model. *Child Development, 55*(1), 83–96.

Bramlett, M. D., & Blumberg, S. J. (2007). Family structure and children's physical and mental health. *Health Affairs, 26*(2), 549–558.

Bronfenbrenner, U., & Morris, P. A. (2006). The bioecological model of human development. In W. Damon & R. M. Lerner (Eds.), *Handbook of child psychology: Theoretical models of human development* (6th ed.). Wiley

Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2019). *Multivariate data analysis* (8th ed.). Cengage Learning.

McCubbin, H. I., & Patterson, J. M. (1983). The family stress process: The double ABCX model of adjustment and adaptation. *Marriage & Family Review, 6*(1–2), 7–37.

Minuchin, S. (1974). *Families and family therapy*. Harvard University Press.

Paulus, F. W., Ohmann, S., Möhler, E., Plener, P., & Popow, C. (2021). Emotional dysregulation in children and adolescents with psychiatric disorders: A narrative review. *Frontiers in Psychiatry, 12*, 628252.